



Yueyang LFG Power Generation Project



Document Prepared by CTI Certification Co., Ltd

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Summary:

CTI Certification Co., Ltd. (hereafter referred to as "CTI") has been commissioned by Climate Bridge (Shanghai) Ltd. to perform the validation and verification of project activity "Yueyang LFG Power Generation Project" (hereafter referred to as "the project activity") and reported in the joint Validation & Verification Report.

The project activity is a landfill gas recovery and power generation project located at the west of Yueyang County MSW Landfill site in Weixing Village, Xinkai Town, Yueyang County, Yueyang City, Hunan province, People's Republic of China, of which the main purpose is to use landfill gas for electricity generation. The total installed capacity of the project activity is 2.134MW, consisting of 2 sets of 1.067MW generators. The project uses LFG from Yueyang County MSW Landfill Site for power generation. During the 10 years fixed crediting period, the power generated by the project is expected to be 48,785 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). Meanwhile by utilization the landfill gas recovered by project, the project activity avoids the methane emissions that would be generated under landfill condition. It's estimated that the project activity could achieve average annual GHG emission reductions of 28,517 tCO₂e during the crediting period.

In the course of the validation and verification, 2 Clarification Requests (CLs) and 1 Corrective Action Request (CAR) was raised and successfully closed. No Forwarded Action Requests (FARs) was raised.

Validation Summary

The validation objective is an independent assessment by a Third Party of a proposed project activity against all defined criteria set for the registration under the VCS. In order to confirm that the project activity, as documented, is sound reasonable and meets the identified criteria, the validation involves the assessment of: project conformance to VCS standards/programs, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the Joint Project Description and Monitoring Report (hereafter refer to as "joint PD & MR") will generate verifiable GHG data and information when implemented. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of VCU. Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for registration to Verra registry.

Validation is conducted using CTI procedures in line with the requirements specified in the latest version of the VCS Validation and Verification Manual and applying auditing techniques. The assessment team assessed the project activity's compliance against the VCS Version 4.4, the selected CDM methodology and the joint PD & MR. The project is eligible under non-AFOLU project. The validation criteria followed the guidance documents provided by VCS and the selected CDM methodology and tool includes:

- CDM Validation and Verification Standard for project activities version 03.0
- CDM Project Standard for project activities version 03.0
- CDM project cycle procedure for project activities version 03.0
- VCS standard v4.4
- VCS guideline v4.3

- AMS-III.G: Landfill methane recovery, Version 10.0
- AMS-I.D: Grid connected renewable electricity generation, Version 18.0
- Emissions from solid waste disposal sites, version 08.1
- Tool to calculate the emission factor for an electricity system, Version 07.0
- Positive lists of technologies, Version 04.0

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified emission reductions (VERs).

A risk-based approach has been followed to perform this validation activity. The review of the joint PD & MR and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews and project owners have provided CTI Certification Co., Ltd (CTI) with sufficient evidence to verify the fulfilment of the stated criteria of VCS.

Verification Summary

CTI has performed the verification of the emission reductions reported for the project activity “Yueyang LFG Power Generation Project” (VCS Project ID: 4349) for the monitoring period 26/11/2021 to 30/04/2023 to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. These emission reductions are claimed as Verified Carbon Units (VCU) under the Verified Carbon Standard (VCS) Version 4.4.

The verification was performed on the basis of VCS Programme Guide Version 4.3 and VCS standard Version 4.4 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues.

In CTI’s opinion, the GHG emission reductions reported for the project in the joint PD & MR (version 3.0 dated 08/10/2023) are fairly stated. The GHG emission reductions were calculated correctly on the basis of approved methodology AMS-III.G “Landfill methane recovery” (Version 10.0) and the monitoring plan contained in the joint PD & MR (version 3.0 dated 08/10/2023).

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report. Hence, CTI is able to certify that the emission reductions from the “Yueyang LFG Power Generation Project” during the period amount to 30,338 tCO₂e.

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1 INTRODUCTION

1.1 Objective

Climate Bridge (Shanghai) Ltd. has commissioned the CTI to carry out the Verified Carbon Standard (VCS) validation of the project "Yueyang LFG Power Generation Project" (VCS ID 4349).

The objective of validation is an independent assessment by a Third Party of a proposed project activity against all defined criteria set for the registration under the VCS. In order to confirm that the project activity, as documented, is sound reasonable and meets the identified criteria, the validation involves the assessment of: project conformance to VCS standards/programs, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of VCUs. Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for registration to Verra registry.

The objective of verification is the periodic independent review and ex-post determination by an accredited Third Party of the monitored reductions in GHG emissions that have occurred as a result of the VCS project activity during a defined verification period. A verification statement is the written assurance that, during a specific period in time, a project activity achieved the emission reductions as verified. For the project activity, the verification was to verify and provide a verification statement of emission reductions reported for the period 26/11/2021 to 30/04/2023.

1.2 Scope and Criteria

The validation scope is defined as an independent and objective review of the joint Project Design and Monitoring Report to validate that (a) the project design is actual, (b) the baseline scenario is correctly defined as per the applied methodology and relate tools, (c) the project is additional, (d) the monitoring plan can be implemented and is transparent and adequate and (e) all data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature. The joint PD & MR is reviewed against the criteria stated in VCS standard Version 4.4 /4/ and the approved baseline and monitoring methodology AMS-III.G, Version 10.0. The validation was based on the requirements VCS Validation and Verification Manual and VCS programs Version 4.3 /5/ and VCS standards Version 4.4 /4/ applying auditing techniques. The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The scope of the verification is to verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan; evaluate the

GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement; verify that reported GHG emissions data is sufficiently supported by evidence. The verification is conducted as per VCS Program Guide (Version 4.3), VCS Standard (Version 4.4), other relevant requirements defined by Verra and the approved methodology AMS-III.G (Version 10.0) applied by the project. The verification shall ensure that reported emission reductions are complete and accurate in order to be verified.

1.3 Level of Assurance

The joint validation and verification report is based on the joint PD & MR, supporting evidences made available to the validator and information collected through performing interviews and during the on-site assessment. The validation conclusion is assured a reasonable level of assurance.

The joint validation and verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. CTI applied a materiality threshold of 2% with respect to omission or misstatements concerning reported quantities as per para 3.10.1 and 4.1.8 (4) of VCS Standard Version 4.4.

1.4 Summary Description of the Project

Project title	Yueyang LFG Power Generation Project
Project Participants	Yueyang Weiteng New Energy Co., Ltd. (Project Owner, host country, P. R. China) Climate Bridge (Shanghai) Ltd. (China)
Location of the project	Yueyang County MSW Landfill Site in Weixing Village, Xinkai Town, Yueyang County, Yueyang City, Hunan province, People's Republic of China. Geographic coordinates: longitude of 113° 11' 24"E, 29° 11' 13"N
Project start date	26/11/2021, the operation start date
Applied Methodology/Version	AMS-III.G, Version 10.0 AMS-I.D, Version 18.0
Scope/Technical Area	Sectoral scope 01: energy industries (renewable-/non-renewable sources) Sectoral scope 13: Waste handling and disposal
VCS crediting period	26/11/2021 to 25/11/2031

The project activity is a landfill gas recovery and power generation project located at Yueyang County MSW Landfill Site in Weixing Village, Xinkai Town, Yueyang County, Yueyang City, Hunan province,

People's Republic of China, of which the main purpose is to use landfill gas for electricity generation. The total installed capacity of the project activity is 2.134MW, consisting of 2 sets of 1.067MW generators. The project uses LFG from Yueyang County MSW Landfill Site for power generation. During the fixed 10 years crediting period, the power generated by the project is expected to be 48,785 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). Meanwhile by utilization the landfill gas recovered by project, the project activity avoids the methane emissions that would be generated under landfill condition. It's estimated that the project activity could achieve GHG emission reductions of 285,176 tCO₂e during the crediting period.

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

A project specific validation and verification plan was developed to guide the validation and verification auditing process to ensure efficiency and effectiveness.

The purpose of the validation and verification is to present a risk assessment for determining the nature and extent of validation and verification procedures necessary to ensure the risk of auditing error is reduced to a reasonable level. According to the ISO14064-3, the criteria are the policy, procedure or requirement used as reference against which evidence is compared. The validation and verification were performed through means of the following three phases in accordance with the requirement of the applied methodology, and the VCS Standard (Version 4.4) and other relevant VCS requirements.

- A desk review of the joint PD & MR and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the joint validation and verification report and statement.

CTI completed a strategic review and risk assessment of the project's activities and processes in order to gain a full understanding of (if applicable) project details; application of methodology; estimated GHG emission reduction and removals; monitoring; safeguards; SDG contributions etc.

CTI also verified the emission reductions through assessing all factors and issues that constitute the basis for emission reductions from the project, including:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

CTI assessed that the reported information in the joint PD & MR are complete and accurate in question. This joint validation and verification report describes the findings of this assessment. The information of the assessment team has been included in below of this report.

2.2 Document Review

The joint PD & MR /1/ and supporting background documents related to the project implementation were reviewed. Documents review was conducted to ensure consistency with and identify any deviation from VCS program requirements. Desk review included an examination of the project design details,

baseline scenario, additionality, ex ante and monitoring data and parameters, and ex ante quantification of GHG emission reductions. The assessment team also used additional documentation by third parties like host party legislation, technical reports referring to the project design details, baseline scenario, additionality, monitoring or to the basic conditions and technical data.

Furthermore, the joint PD & MR /1/ and emission reduction calculation spreadsheet of monitoring period from 26/11/2021 to 30/04/2023, relevant decisions, clarifications and guidance from the Verra, other information and references relevant to the project activity have also been assessed. During the desk review,

CTI has applied standard auditing techniques to assess the quality of information provided, including a review of the data and information presented to verify their completeness; a review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and an evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

The references used in the course of this validation and verification are summarized in Appendix 3. The validation and verification were performed basing on the documents check and site inspection/measurements, refer to the section 3 of this report for the validation process detail, section 3 of this report for the verification process detail, and Appendix 3 for corresponding documents review.

According to the sectoral scopes / technical area and experiences in the sectoral or national business environment, CTI has composed a project assessment team in accordance with the appointment rules in CTI. The composition of CTI assessment team has to be approved by the CTI ensuring that the required skills are covered by the team. The four qualification levels for team members that are assigned by formal appointment rules as below:

Function	Name	Technical competence	Task Performance*
Team Leader	Feng Tian	1, 8, 13	☒ DR ☒ SV ☒ RP ☐ TR
Technical Reviewer	Lin Shunrong	1, 3, 13, 14, 15	☒ DR ☐ SV ☐ RP ☒ TR

*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

2.3 Interviews

The objective of the interview process was to solicit important information from personnel related to project and relevant to the validation and verification process. Onsite interviews and information discussions were conducted with PP. The key personnel interviewed are summarized in the table below:

Interviewed personnel	Role	Organization	Subject
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Mr. Qin Chuan	Plant Manager	Yueyang Weiteng New Energy Co., Ltd.	- Status of the project (including PPs); - Operation of the project activity, including project technology and equipment etc; - Applicability of selected methodology; - Baseline of the project; - Emission factors; - Applicable laws, statutes and other regulatory frameworks; - The process of the stakeholder consultation; - The impact of the project activity; - The complaint by local stakeholders and the implementation of the mitigation measures; - SDG contribution of the project activity; - Monitoring plan and implementation of the monitor plan of the project activity; - Data collection, QA/QC and achievement; - Training provided; - Calibration of meters and equipment maintenance.
Mr. Fang Qin	Staff		
Mr. Liu Meide	Staff		
Ms. Wang Jing	Staff	Yueyang County MSW Landfill Site	- Status of Yueyang County MSW Landfill Site; - Operation of Yueyang County MSW Landfill Site.
Mr. Li Fan	Staff	Yueyang County Branch of Yueyang Ecology and Environment Bureau	- The process and participation of the stakeholder consultation; - The impact of the project activity; - The complaint by local stakeholders and the implementation of the mitigation measures; - SDG contribution of the project activity.
Mr. Zhang Jianguo	Villager	Local Village	
Mr. Gu Zhenguo	Villager		
Ms. Li Qing	Villager		
Ms. Yang Xiaoyu	Villager		
Mr. Jiang Jikang	Project Manager	Climate Bridge (Shanghai) Ltd.	Data collection and ER calculation.

2.4 Site Visits

The assessment team performed the on-site validation and verification (Yueyang County MSW Landfill Site in Weixing Village, Xinkai Town, Yueyang County, Yueyang City, Hunan province, People's Republic of China) on 28/05/2023 - 29/05/2023. The interviewed personnel and objective are listed in above table.

2.5 Resolution of Findings

As an outcome of the validation process, the team can raise different types of findings.

Where a non-conformance arises the assessment team shall raise a Corrective Action Request (CAR). A CAR is issued, where:

- a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient;
- b) Modifications to the implementation, operation and monitoring of the project activity has not been sufficiently documented by the project participants;
- c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;
- d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

The assessment team shall raise a Clarification Request (CL) if information is insufficient or not clear enough to determine whether the applicable CDM or VCS requirements have been met.

All CARs and CLs raised during validation shall be resolved prior to submitting a request for issuance.

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for CTI's positive conclusion on the joint PD & MR /1/. 2 Clarification Requests (CLs), 1 Corrective Action request (CAR) and no Forward action request (FAR) raised by CTI. And communications were made between the Client and CTI to guarantee the transparency of the validation and verification process, the concerns raised, and responses given are summarized below in the appendix 4.

The final joint PD & MR (version 3.0 dated 08/10/2023) serves as the basis for the final assessment presented. Additional changes to the project during the validation and verification process are not considered to be significant with respect to the main CDM/VCS objectives. The two CDM/VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation and verification findings	No. of CL	No. of CAR	No. of FAR
Project Description and monitoring report	1	00	00
Description of project activity	00	00	00
Application of selected baseline and monitoring methodology and selected standardized baseline			
- Applicability of methodology and standardized baseline	00	00	00
- Deviation from methodology	00	00	00
- Clarification on applicability of methodology, tool and/or standardized baseline	00	00	00
- Demonstration of additionality	00	00	00
- Emission reductions	00	00	00
- Monitoring plan	00	00	00
- Stakeholders consultation process	00	00	00

- Public comments	00	00	00
Others (please specify) - Matter related to double counting - for validation	00	00	00
Others (please specify) - Matter related to project operation - for verification	1	00	00
ER achieved, Matter related to feeder details, breakdown and Calibration - for verification	00	1	00
Total	2	1	00

2.5.1 Forward Action Requests

None FAR was raised during the validation process.

3 VALIDATION FINDINGS

3.1 Project Details

Project type, technologies and measures implemented, and eligibility of the project

The project activity is a landfill gas recovery and power generation project of which the main purpose is to use landfill gas for electricity generation. The total installed capacity of the project activity is 2.134MW, consisting of 2 sets of 1.067MW generators.

The project uses LFG from Yueyang County MSW Landfill Site for power generation. During the 10 years fixed crediting period, the power generated by the project is expected to be 48,785 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). Meanwhile by utilization the landfill gas recovered by project, the project activity avoids the methane emissions that would be generated under landfill condition. It's estimated that the project activity could achieve average annual GHG emission reductions of 28,517 tCO₂e during this crediting period.

The project activity is located at Yueyang County MSW Landfill Site in Weixing Village, Xinkai Town, Yueyang County, Yueyang City, Hunan province, People's Republic of China. The project activity has been developed by Yueyang Weiteng New Energy Co., Ltd. The geographic coordinates of the project activity are 113° 11'24"E, 29° 11'13"N which is confirmed by site visit.

By checking the Project Approval /16/, the assessment team confirmed that the Project activity has been approved by Yueyang County Development and Reform Bureau on 29/06/2020. By checking Environmental Impact Assessment (EIA) approval /18/, the assessment team confirmed that the project activity has been approved by Yueyang County Branch of Yueyang Ecology and Environment Bureau on 11/11/2020. By checking the Construction contract signed between the project owner and Yueyang City Daming Construction Co., Ltd /21/, the assessment team confirmed that the project started construction on 02/12/2020. By checking Grid Connection Acceptance Opinion of the project

activity issued by local power grid company and site interview, the assessment team confirmed that the project activity started operation on 26/11/2021.

For the project activity, the applicability of VCS program as required in para 2.1.1 of VCS Standard (version 4.4) is justified as below:

Criteria	Justification
The scope of the VCS Program includes: 1) The seven Kyoto Protocol greenhouse gases.	During the 10-year fixed crediting period, the Project can reduce CO ₂ emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). Meanwhile by utilization the landfill gas (mainly methane) recovered by project, the project activity avoids the methane emissions that would be generated under landfill condition. Therefore, this criteria is applicable.
2) Ozone-depleting substances.	This criteria is not applicable for the project activity.
3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	This criteria is not applicable for the project activity.
4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	The project activity applies methodology AMS-III.G. (version 10.0) and AMS-I. D. (version 18.0), the methodologies approved under CDM Program, which is a VCS approved GHG program.
5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.	This criteria is not applicable for the project activity.
The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction. The VCS Program also excludes the following project activities under the circumstances indicated in Table 1 of VCS Standard (version 4.4).	The Project utilized the captured landfill gas (mainly methane) from Yueyang County MSW Landfill Site for electricity generation. The main purpose is to avoid the methane emissions that would be generated under landfill condition and to reduce CO ₂ emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). The project activity is not excluded under the circumstances indicated in Table 1 of VCS Standard (version 4.4).

Project design, including eligibility criteria for grouped projects

By checking Technical Specifications /19/, nameplates of gas generators /20/ and public information /39/ during site visit, it was verified the technical parameters of gas generators were:

Manufacturer	JENBACHER
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Type	JGS320GS-B.L.
Quantity	2
Rated capacity	1.067 MW
Rated power factor	1
Rated frequency	50 HZ
Rated voltage	400 V
Rated current	1,540 A
Lifetime	20 years

The project activity is not a grouped project activity. Therefore, this section is not applicable.

Project proponent and other entities involved in the project

Through document review and site interview, the assessment team confirmed the details of the project proponent as:

Organization name	Yueyang Weiteng New Energy Co., Ltd.
Contact person	CHENG Xuan
Title	Manager
Address	Room 3102, Building 3, Dongfang Road, Rongjiawan Town, Yueyang Town, Yueyang County, Yueyang City, Hunan province, China
Telephone	+86 -2123019950
Email	3542346576@qq.com

Organization name	Climate Bridge (Shanghai) Ltd.
Role in the project	Consultancy
Contact person	Gao Zhiwen
Title	General Manager
Address	Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, China 200120
Telephone	+86-21 62462036
Email	projects@climatebridge.com

Project Ownership

By checking business license /14/, Project Approval /16/ and EIA Approval /18/, the assessment team confirmed that Yueyang Weiteng New Energy Co., Ltd. is the project owner and also project proponent (PP) of project activity. Therefore, Yueyang Weiteng New Energy Co., Ltd. has the legal right to control and operate the project activities.

Project Start Date

By checking Operation Log /35/ of the project activity and site interview, the assessment team confirmed that the project activity started operation on 26/11/2021, which is the start date of the project activity.

Project crediting period Date

The project activity adopts 10-year fixed crediting period. the assessment team confirms that the starting and ending dates of crediting period is as below:

Crediting Period Start date: 26/11/2021

Crediting Period End date: 25/11/2031

Project Scale and Estimated GHG Emission Reductions or Removals

The total installed capacity of the project activity is 2.134MW consisting of 2 sets of 1.067MW generators. As the estimated annual average GHG emission reductions or removal per year is 28,517 tCO₂e which is less than 300,000 tonnes of CO₂e per year, thus the project falls in the category of Project. Therefore, the assessment team confirms that the project scale falls under Project.

Project Scale	
Project	✓
Large project	

Through checking emission reductions calculation spreadsheet provided by PP, the assessment team was able to confirm that the estimated GHG Emission Reductions or Removals of the project activity is as follows:

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
26/11/2021 to 31/12/2021	5,059
2022	48,817
2023	42,097
2024	36,421
2025	31,620
2026	27,550
2027	24,094
2028	21,154
2029	18,646
2030	16,501
01/01/2031 to 25/11/2031	13,217
Total estimated ERs	285,176
Total number of crediting years	10

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Average annual ERs	28,517

The above estimated emission reduction is confirmed by the assessment team via emission reduction calculation spreadsheet. the assessment team confirmed that the calculation is correct and conservative.

Project location

The project activity is located at Yueyang County MSW Landfill Site in Weixing Village, Xinkai Town, Yueyang County, Yueyang City, Hunan province, People's Republic of China. The project activity has been developed by Yueyang Weiteng New Energy Co., Ltd.. The project is located at 113°11'24"E, 29°11'13"N which is confirmed by site visit.

Conditions prior to project initiation

Before the implementation of the project activity, the electricity generated by the project would be supplied by CCPG in the baseline scenario and the landfill gas combusted would be disposed in the landfill without utilization.

Project compliance with applicable laws, statutes and other regulatory frameworks

The assessment team confirms that the Project has been approved by Chinese government by checking the Project approval /16/ and Environmental Impact Assessment (EIA) approval /18/.

By checking laws and regulation, it is confirmed that the project activity is in complicate with all laws and regulations in China.

Participation under other GHG programs

Projects registered (or seeking registration) under other GHG program(s): The project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program. A declaration from PP has been provided and checked by CTI.

Also, the assessment team checked the registries of CDM EB and Goldstandard through website <http://cdm.unfccc.int/> and <http://www.goldstandard.org/> to confirm the same.

Rejection by other GHG programs: The Project is not rejected by other GHG programs. A declaration for the same has been provided and checked by CTI.

Also, the assessment team checked the registries of CDM EB, Goldstandard and Verra through website <http://cdm.unfccc.int/>, <http://www.goldstandard.org/> and <http://verra.org/> to confirm the same.

Other forms of credit and supply chain (Scope 3) emissions

Emissions trading programs and other binding limits: The assessment team confirms that the net GHG emission reductions or removals generated by the Project will not be used for compliance with an

emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits. The assessment team checked the REC Mechanism database of China and Chinese Emission Trading System (Chinese ETS) and found that the project activity is not accredited / registered under REC mechanism or Chinese ETS. Further, Declaration in effect of the same has been submitted by project proponent to the assessment team and found to be correct. Thus, the assessment team concluded that the project activity not involved on other Emissions trading programs and other binding limits.

Furthermore, as per "Notice on Strengthening Enterprise Greenhouse Gas Emission Report Management" /30/ issued by National Development and Reform Committee of P. R. China, China has a national emissions trading scheme only cover the high-emission industries, including thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO₂e/year. By checking "List of key emissions in the national carbon emissions transaction quota management in 2019-2020" /31/ issued by Ministry of Ecology and Environment of P.R.China, the project activity is not included in the mandatory emission control scheme and thus no emission cap was enforced for the project activity.

Other forms of environmental credit sought or received and eligible to be sought or received: The Project has no intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

Renewable energy certificates are available for trading in the host country. However, the same is not availed by the project proponent. The undertaking regarding the same is submitted by PP which is acceptable to the assessment team. the assessment team also checked the REC website and found the declaration to be correct.

Issuance of public statement(s) to help prevent Scope 3 emissions double claiming: The Project can reduce GHG emissions by utilization the recovered landfill gas provided by Yueyang County MSW Landfill Site, avoiding methane emissions that would be released into atmosphere; and by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). As confirmed with PP and interview during site visit, only VCUs will be claimed by PP, and Yueyang County MSW Landfill Site has been orally informed to avoid double counting. Besides, as per VCS standard version 4.4, public statements are required for all emission reductions associated with renewable electricity for which VCUs may be requested from 01/07/2023 onwards. Therefore, the assessment team confirmed that this section is not applicable for the project activity.

Email notification of the potential risk of Scope 3 emissions double claiming: The Project can reduce GHG emissions by utilization the recovered landfill gas provided by Yueyang County MSW Landfill Site, avoiding methane emissions that would be released into atmosphere; and by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). As confirmed with PP and interview during site visit, only VCUs will be claimed by PP, and Yueyang County MSW Landfill Site has been orally informed to avoid double counting. Besides, as per VCS standard version 4.4, public statements are required for all emission reductions associated with renewable electricity for which

VCUs may be requested from 01/07/2023 onwards. Therefore, the assessment team confirmed that this section is not applicable for the project activity.

Additional information relevant to the project

Leakage management for AFOLU projects: The project activity is not a AFOLU project, therefore, this section is not applicable.

Commercially sensitive information: No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the assessment team for analysis which lead to positive conclusion for this validation.

Sustainable development contributions: The project activity would contribute sustainable development in the region in following aspects confirmed by site visit:

By checking EIA Report /17/, Staff Payroll Proof /43/, China's National Plan on Implementation of the 2030 Agenda for Sustainable Development, China's 14th Five-Year Plan and the 2035 Vision Outline /37/, and 17 SDGs defined by UNDP, interviewing stakeholders during site visit, the validation team confirmed that the project activity would contribute sustainable development in the region and China's Sustainable Development Goals (SDGs). The specific sustainable development goals that the project activity could achieve have been listed as follows:

- **SDG13 "Take urgent action to combat climate change and its impacts":** The project recovers and destroys landfill gas that consists mainly of greenhouse gas methane and would otherwise be released directly into the atmosphere, reducing air pollution and methane emissions to the surrounding area, decreasing odour nuisance and minimizing health risks of local residents, and increasing the safety on the landfill. Proper extraction and utilization of the landfill gas will minimize the potential for LFG migration on site, thus reducing the explosion risks of landfill gas. This contributes to achieve one of China's stated sustainable development priorities "China's carbon dioxide emissions would strive to peak by 2030 and strive to achieve carbon neutrality by 2060".

During this monitoring period, the emission reductions achieved were 30,338 tCO₂e and the cumulative Contributions Over Project Lifetime since project start date were 30,338 tCO₂e achieved emission reductions.

- **SDG 7 "Ensure access to affordable, reliable, sustainable and modern energy for all":** The project provides clean and renewable energy source and helps to improve energy supply security; displacing the power generation of fossil fuel power plants by using renewable energy sources, reducing fossil fuel consumption in Central China Power Grid and further helping to achieve the national action to promote renewable energy development. This contributes to achieve one of China's stated sustainable development priorities "By 2030, the proportion of non-fossil energy consumption will reach about 25%".

During this monitoring period, the renewable energy supplied to CCPG was 5,675.868 MWh and the cumulative Contributions Over Project Lifetime since project start date were 5,675.868 MWh renewable energy supplied to CCPG.

- **SDG 8 "Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all"**: the project activity will increase employment opportunities for the operation of the project activity. This contributes to one of China's actions for promoting sustainable developing: "by 2030, achieve full and productive employment and decent work for all women and men, including young people and persons with disabilities, and equal pay for work of equal value".

By checking Staff Payroll Proof /43/, it is confirmed that since operation of the project activity (project start date), 12 long term working opportunities are provided.

3.2 Participation under Other GHG Programs

The project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program. The Project is not rejected by other GHG programs. A declaration from PP has been provided and checked by CTI.

Also, the assessment team checked the registries of CDM EB and Goldstandard through website <http://cdm.unfccc.int/> and <http://www.goldstandard.org/> to confirm the same.

Therefore, the assessment team confirmed that the joint PD & MR is accurate, complete, and provides an understanding of the nature of the project.

3.3 Safeguards

3.3.1 No Net Harm

The Environmental Impact Assessment (EIA) Report of the project activity was compiled by Hunan Huanmeida Environmental Protection Technology Co., Ltd. and approved by the Yueyang County Branch of Yueyang Ecology and Environment Bureau on 11/11/2020. The assessment team confirm all environmental impacts has been analyzed and no net harm was detected. Refer to section 3.3.3 of this report for detailed environmental impacts arising from the Project construction and operation.

Through interview with local stakeholders, it is confirmed by the assessment team the implementation of the project will improve local socio-economic development and contribute to the sustainable development as described in section 3.2 of this report above.

In conclusion, it is confirmed by the assessment team the project has no negative impacts on local environment and socio-economy. And no net harm on local environment and social community has been detected for the project.

3.3.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to invite the relevant stakeholders, prior of the validation process. The assessment team checked the relevance of the dates during the validation site visit.

Local stakeholder consultation was conducted during EIA stage as per national regulation “Interim Measures for public participation in environmental impact assessment” (Huanfa [2006] No.28) published by The Ministry of environmental protection of P.R. China. Survey questionnaires were distributed to relevant personnel of the landfill site, local villagers, and government officials by the Project owner in November 2020. The questionnaire was reasonably designed to assess the project impacts on the local environment and social economic development. In total 50 out of 50 questionnaires were returned with a 100% response rate. By checking the questionnaires, the assessment team confirm that the local stakeholder has no negative comments for the construction of the project activity.

Furthermore, through site interview, the assessment team confirmed that communications with stakeholders were conducted at periodic intervals during project implementation period. Key implementation schedules or changes of the project has been and will be communicated to the local authority, neighbourhood committee and local residents. Local government agencies and authorities will conduct spot checks on the implementation of the project, and give suggestions to the project owner on the involved rectification problems. By checking the filled questionnaires from local stakeholders /36/, the assessment team confirmed that communications with local stakeholders was being carried out in 23/05/2022 and 16/01/2023 by the project owner during this monitoring period. There are no negative comments received for the project during this monitoring period.

Furthermore, via interviewing with staff from local Ecology and Environment Bureau during the site visit, it is confirmed that during the implementation stage of this monitoring period, local authority has conducted spot checks and the project passed all the periodic spot checks by local government.

Therefore, the assessment team was able to confirm that the stakeholder meeting was adequate and appropriate.

3.3.3 Environmental Impact

An Environmental Impact Assessment has been conducted by the project participants. The assessment team has reviewed the documentation of the presented information, eg. EIA Report /17/ and EIA Approval /18/. The EIA Approval confirms the correctness of the approach used by PPs. In conclusion, the PPs have followed the requirements of the host country with regards to addressing environmental impacts as below:

Air Pollution

By checking EIA Report /17/ and site interview with stakeholders, the assessment team confirmed during the construction period, construction dust and road dust has a certain effect to the surrounding areas of construction site. Besides, a small amount of malodorous gas has generated by drilling. Effective measures, i.e. sprinkling water to prevent floating dust, prohibiting on-site mixing of concrete, covering construction slag piles, cleaning transport vehicles regularly, and strengthening environmental

education for construction workers, are taken to prevent impacts on construction and road dust. Besides, effective measures, i.e. partitioning construction, recommending construction workers to wear masks and spraying a small amount of deodorant, are taken to prevent impacts on malodorous gas. Also, the management has been strengthened to limited the impacts and the influence has been disappeared once the construction was finished.

During the operation period, the project uses LFG which is collected from Yueyang County MSW Landfill Site to generate electricity, avoiding uncontrolled releasing of LFG. By checking EIA Report /17/ and site inspection, it was verified that standardize the construction of landfill gas pretreatment devices have been installed. The landfill gas will enter the generator unit after denitration, purification and other pretreatment. Therefore, it reduces greenhouse gas and effluvium emitting to air, and mitigates the possible danger of fire or explosion. By checking EIA Report /17/ and site visit, the assessment team confirmed the exhaust gas produced by the project is mainly the exhaust gas of the generators. The exhaust gas from each generator is respectively discharged through exhaust cylinders with 8m high. The emission of NO_x, SO₂ and particulate matter in the exhaust gas satisfied with relevant national standards.

Therefore, the emission of exhaust gas has little impact on surrounding environment.

Wastewater

During the construction period, wastewater mainly includes construction wastewater and domestic sewage. By checking EIA Report /17/ and site interview with stakeholders, the assessment team confirmed domestic sewage is discharged into the leachate treatment tank of Yueyang County MSW Landfill Site for treatment; the construction wastewater was reused after oil separation and sedimentation.

During the construction period, the wastewater is mainly domestic sewage and condensate. By checking the EIA Report /17/ and site visit, the assessment team confirmed the domestic sewage and condensate will be treated by the leachate treatment station of Yueyang County MSW Landfill Site before discharging to municipal sewage treatment plant for further treatment, which satisfies with relevant national standard. Furthermore, the sealing and anticorrosion of sewage pipe network and the anti-seepage and leakage prevention of hazardous waste temporary storage sites has been done to prevent the leakage of liquid from polluting the groundwater in the area, which is based on the principle of partition prevention and control. During site inspection, it was verified the rainwater enters the rainwater pipeline of the landfill site through the rainwater pipeline of the plant area, and finally enters the Xinqiang River.

Noise

By checking the EIA Report /17/, the assessment team confirmed that the noise generated by the project construction has a slight impact on the surrounding sensitive points, and its pollution impact is localized and short-term. Reasonable measures, i.e. use low-noise operating equipment, optimize the construction layout, strictly control noise, have been taken to reduce the noises to acceptable range.

The noise meets the national standard Emission Standard of Environment Noise for Boundary of Construction Site (GB12523-2011).

During the operation period, the noise of this project comes from the production process, various mechanical operation and vibration. As per the EIA Report /17/ and site visit, the assessment team confirmed that measures, i.e. shock absorption, sound insulation and noise elimination, have been taken to effectively reduce the noise at the factory boundary. The noise meets Category 2 requirements in the national standard Emission Standard for Industry Enterprise Noise at Boundary (GB 12348-2008).

Solid Waste

During construction period, the solid waste construction waste and domestic waste. By checking the EIA Report /17/ and interview with stakeholder, the assessment team confirmed during the construction period, construction waste was cleaned and buried in Yueyang County MSW Landfill Site while domestic waste was collected and land-filled in Yueyang County MSW Landfill Site.

During the operation period, the solid waste of the project is mainly domestic garbage, waste engine oil catalyst, waste adsorbent and waste filter element. By checking the EIA Report /17/ and site visit, the assessment team confirmed the domestic garbage are collected uniformly by environmental sanitation department. The waste adsorbent and waste filter element will be recycled by the manufacturer regularly. Hazardous waste, eg. waste engine oil and catalyst will be properly collected and stored in the hazardous waste temporary deposit and will be collected regularly by qualified entities for centralized treatment.

In conclusion, after the above measures are performed, the negative impacts on environment will be minimized below the requirements of laws and regulations during the construction and operational period.

3.3.4 Public Comments

The assessment team noted that this project was open for public comment from 03/05/2023 to 02/06/2023. The detail was checked by the assessment team in the following web platform: <https://registry.verra.org/app/projectDetail/VCS/4349>.

During the period, no public comments were received.

3.3.5 AFOLU-Specific Safeguards

The project activity is not an AFOLU project.

For non-AFOLU projects, this section is not required.

3.4 Application of Methodology

3.4.1 Title and Reference

The assessment team checked that following methodology and tools are applicable for the project activity as below:

Title	Landfill methane recovery, AMS-III.G, Version 10.0 Grid connected renewable electricity generation, AMS-I.D., Version 18.0
Reference	The project activity meets the eligibility criteria of small-scale project as: - the estimated annual average GHG emission reductions or removal per year is 28,517 tCO ₂ e which is less than 60,000 tonnes of CO ₂ e per year; - the installed capacity is 2.134 MW which is less than 15 MW.
Methodology	Landfill methane recovery, AMS-III.G, Version 10.0 Grid connected renewable electricity generation, AMS-I.D., Version 18.0
Type	Energy industries (renewable / non-renewable sources)
Sectoral scope(s)	01 and 13
Category	Approved Consolidated Methodology
Tools	Tools referred with above methodology and applicable for project activity are: - Emissions from solid waste disposal sites, version 08.1 - Tool to calculate the emission factor for an electricity system, Version 07.0 - Positive lists of technologies, Version 04.0

3.4.2 Applicability

The applied baseline methodology is justified as it has been demonstrated that the proposed project activity is:

Applicability conditions of "Landfill methane recovery"

Applicability criteria	Justification of the project situation
This methodology comprises measures to capture and combust methane from landfills (i.e. solid waste disposal sites) used for the disposal of residues from human activities including municipal, industrial, and other solid wastes containing biodegradable organic matter.	By checking EIA Report /17/, FSR /15/ and through on-site interview, the assessment team confirmed that the project activity comprises measures to capture and combust methane from Yueyang County MSW Landfill Site used for the disposal of residues from human activities including municipal, industrial, and other solid wastes containing biodegradable organic matter. Therefore, the assessment team was able to confirm that the methodology is applicable.

Different options to utilise the recovered landfill gas as detailed in paragraph 4 of “AMS- III.H.: Methane recovery in wastewater treatment” (version 19.0) are eligible for use under this methodology. The relevant procedures in AMS-III.H. shall be followed in this regard. As per para. 4 of AMS- III.H., the recovered biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case additional guidance provided in the appendix shall be followed;or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided in the appendix shall be followed: (i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or (iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users; (d) Hydrogen production; (e) Use as fuel in transportation applications after upgrading.	By checking EIA Report /17/, FSR /15/ and through on-site interview, the assessment team confirmed that the project activity captures and utilizes landfill gas for electrical energy generation directly, which satisfied with (a) of para. 4 of AMS-III.H. Therefore, the assessment team was able to confirm that the methodology is applicable.
Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO₂ equivalent annually from all Type III components of the project activity.	The estimated annual average GHG emission reductions or removal per year of the project activity is 28,517 tCO₂e, which is less than 60 kt CO₂e annually from all Type III components of the project activity. Therefore, the assessment team was able to confirm that the methodology is applicable.
The proposed project activity does not reduce the amount of organic waste that would have been recycled in the absence of the project activity.	By checking EIA Report /17/, FSR /15/, Operation Management Manual of Yueyang County MSW Landfill Site /25/ and through on-site interview, the assessment team confirmed that the project does not impact the management of the landfill, and will not reduce the amount of organic waste that would have been recycled in the absence of the project

	activity. Therefore, the assessment team was able to confirm that the methodology is applicable.
This methodology is not applicable if the management of the solid waste disposal site (SWDS) in the project activity is deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g. other than to meet a technical or regulatory requirement). Such changes may include, for example, the addition of liquids to a SWDS, pre-treating waste to seed it with bacteria for the purpose of increasing the rate of anaerobic degradation of the SWDS or changing the shape of the SWDS to increase methane production.	By checking EIA Report /17/, FSR /15/, Operation Management Manual of Yueyang County MSW Landfill Site /25/ and through on-site interview, the assessment team confirmed that the management of the SWDS in the project activity will not be deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity. Therefore, the assessment team was able to confirm that the methodology is applicable.

Applicability conditions of "Grid connected renewable electricity generation"

Applicability criteria	Justification of the project situation
This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	By checking EIA Report /17/, FSR /15/ and through on-site interview, the assessment team confirmed that the project utilizes LFG to supply electricity to Central China Power Grid. Therefore, the assessment team was able to confirm that the methodology is applicable.
This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s).	By checking EIA Report /17/, FSR /15/ and through on-site interview, the assessment team confirmed that the project installs a Greenfield LFG power plant at the existing Yueyang County MSW Landfill Site, where there was no renewable energy power plant operating prior to the implementation of the project activity. Therefore, the assessment team was able to confirm that the methodology is applicable.
Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is	By checking EIA Report /17/, FSR /15/ and through on-site interview, the assessment team confirmed that the project activity is not a hydro power plant. Therefore, this criteria is not applicable.

increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m²; (c)The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m².	
If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	By checking EIA Report /17/, FSR /15/ and through on-site interview, the assessment team confirmed that the new unit has only renewable components and the total installed capacity of the project activity is 2.134MW, consisting of 2 sets of 1.067MW generators. The installed capacity. The total installed capacity of the project activity is less than 15 MW, therefore, the project activity is below the eligibility limit for a small-scale project activity.
Combined heat and power (co-generation) systems are not eligible under this category.	By checking EIA Report /17/, FSR /15/ and through on-site interview, the assessment team confirmed that the project activity does not involve combined heat and power (co-generation) systems. Therefore, this criteria is not applicable.
In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	By checking FSR /15/ and through on-site interview, the assessment team confirmed that the project activity does not involve the capacity addition at an existing renewable power generation facility. Therefore, this criteria is not applicable.
In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	By checking EIA Report /17/, FSR /15/ and through on-site interview, the assessment team confirmed that the project does not involve retrofit, rehabilitation, or replacement. Therefore, this criteria is not applicable.
In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored.	By checking EIA Report /17/, FSR /15/ and through on-site interview, the assessment team confirmed that the project recovers LFG for electricity generation. As justified above, the project is eligible under a relevant Type III category. Since the recovered methane is used for electricity generation for supply to CCPG, then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology.

In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply.	By checking EIA Report /17/, FSR /15/ and through on-site interview, the assessment team confirmed that the project does not involve biomass sourced from dedicated plantations. Therefore, this criteria is not applicable.
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Applicability conditions of "Emissions from solid waste disposal sites"

Applicability criteria	Justification of the project situation
Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. "ACM0001: Flaring or use of landfill gas"). The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (CDM-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g. measuring the amount of methane captured from the SWDS); Application B: The CDM project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions. These two types of applications are referred to in the tool for determining parameters.	By checking the EIA Report /17/, FSR /15/ of the project activity, the assessment team confirmed that the project activity mitigates methane emissions from Yueyang County MSW Landfill Site. Methane emissions are mitigated by capturing for power generation. The methane is generated from waste disposed in the past, including prior to the start of the project activity. The tool is only applied for an ex ante estimation of emissions in the project description. The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g. measuring the amount of methane captured from the SWDS). Therefore, the assessment team confirmed that the project adopts Application A.
In the case that: (a) different types of residual waste are disposed or prevented from disposal; or that (b) both MSW and residual waste(s) are prevented from disposal, then the tool should be applied separately to each residual waste and to the MSW.	By checking the EIA Report /17/, FSR /15/ and through on-site interview, the assessment team confirmed that only MSW are disposed in Yueyang County MSW Landfill Site.

Applicability conditions of "Tool to calculate the emission factor for an electricity system"

Applicability criteria	Justification of the project situation
This tool may be applied to estimate the OM, BM	By checking the PPA /23/ of the project activity,

and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	the assessment team confirmed that the electricity supplied by the project was exported to CCPG. OM, BM and CM are estimated using the tool for calculating baseline emissions for the project activity.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	By checking the PPA /23/ of the project activity, the assessment team confirmed that the electricity generated by the project activity was exported to CCPG. The emission factor for the project electricity system is calculated for grid power plants.
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	By checking the EIA Approval /18/, Project Approval /16/ of the project activity, the assessment team confirmed the project activity is located in P. R. China, not in any Annex I country.
Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	The project activity does not involve biofuels.

Applicability conditions in tool "Positive lists of technologies"

Applicability criteria	Justification of the project situation
The use of this methodological tool is not mandatory for the project participants of a CDM project activity or CDM PoA for demonstrating their additionality.	The project activity adopts this methodological tool for demonstrating the additionality.
This methodological tool shall be applied in conjunction with a small-scale or small-scale methodology which refers to this tool.	The methodological tool is applied in conjunction with AMS-III.G which refers to this tool.
The positive lists as contained in section 5 of this tool are valid up to 10/03/2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and	The positive lists as contained in section 5 "5.1.1. Landfill gas recovery and its gainful use" of this tool are valid up to 10/03/2025.

methodological tools in the "Procedure: Development, revision and clarification of baseline and monitoring methodologies and methodological tools", there will be no grace period for the application of this tool and the validity of the positive list after this date, including in cases where further technologies are added to the positive list through revisions of this tool before this date.	
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The assessment team confirmed that the application of the baseline methodology is transparent and conservative and confirms that the chosen baseline and monitoring methodology i.e. AMS-III.G, Version 10.0 is applicable to the project activity.

The project activity qualifies as Type III during every year of the crediting period in accordance with applicable provisions for project activity eligibility as discussed above. Also, annual average GHG emission reductions or removal per year of the project activity is 28,517 tCO₂e, less than 60,000 tonnes of CO₂e per year. Therefore, the project is applicable as per small scale project activities methodology AMS-III.G Version 10.0.

3.4.3 Project Boundary

As per AMS-III.G, the project boundary of the project activity shall include the physical, geographical site of the landfill where the gas is captured and destroyed/used.

The assessment team confirmed that the project boundary of the project activity includes the sites where the LFG is flared or used (e.g. power plant). The electricity used by the project activity is sourced from CCPG and electricity generated by LFG capturing of the project activity is connected to CCPG. Therefore, the project boundary includes the whole LFG related system (e.g. LFG collection, LFG pre-treatment system, LFG power generation system, flare system, etc.) and all grid-connected power plants in CCPG. According to China DNA [http://www.mee.gov.cn/ywgz/ycqhbh/wsqtgz/202012/W020201229610353340851.pdf], CCPG covers Henan Province, Hubei Province, Hunan Province, Jiangxi Province, Sichuan Province and Chongqing City.

The sources and GHG gases involved for the Project activity are as below.

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition of waste at the SWDS site	CO ₂	No	CO ₂ emissions from decomposition of organic waste are not accounted since the CO ₂ is also released under the project activity.
		CH ₄	Yes	The major source of emissions in the baseline
		N ₂ O	No	N ₂ O emissions are small compared to CH ₄ emissions from SWDS. This is conservative.
	Emissions from electricity generation	CO ₂	Yes	Major emission source
		CH ₄	No	Excluded for simplification. This is

Source		Gas	Included?	Justification/Explanation	
				conservative.	
		N ₂ O	No	Excluded for simplification. This is conservative.	
	Emissions from heat generation	CO ₂	No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no heat generation is included in the project activity.	
		CH ₄	No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no heat generation is included in the project activity.	
		N ₂ O	No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no heat generation is included in the project activity.	
	Emissions from the use of natural gas	CO ₂	No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity.	
		CH ₄	No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included.	
		N ₂ O	No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity.	
	Project activity	Emissions from fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity	CO ₂	No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no fossil fuel is used by the project activity.
			CH ₄	No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no fossil fuel is used by the project activity.
N ₂ O			No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no fossil fuel is used by the project activity.	
Emissions from electricity consumption due to		CO ₂	Yes	May be an important emission source	
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.	

Source		Gas	Included?	Justification/Explanation
	the project activity	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
	Emissions from flaring	CO ₂	No	Emissions are considered negligible
		CH ₄	No	Through site interview with staff from Yueyang County MSW Landfill Site operation entity, the assessment team confirmed that the flare will only be involved when the electricity generation system is not in operation. As confirmed with PP, VERs from flare will not be claimed, which is conservative. Under this situation, no baseline emission from flaring will be calculated and it is reasonable the project emissions from flaring are excluded. And Project emissions from flaring system are excluded for simplification.
		N ₂ O	No	Emissions are considered negligible
	Emissions from distribution of LFG using trucks and dedicated pipelines	CO ₂	No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no distribution of LFG using trucks and dedicated pipelines is involved in the project activity.
		CH ₄	No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no distribution of LFG using trucks and dedicated pipelines is involved in the project activity.
		N ₂ O	No	By checking EIA Report /17/, FSR /15/ and through on-site interview and inspection, it was verified no distribution of LFG using trucks and dedicated pipelines is involved in the project activity.

3.4.4 Baseline Scenario

As per para. 13 of AMS-III.G, if the LFG is used for heat and electricity generation within the project boundary, that component of the project activity may use a corresponding methodology under Type I project activities: If all or part of the electricity generated by the project activity is exported to the grid, the baseline scenario for all or the part of the electricity exported to the grid is assumed to be electricity generation in existing and/or new grid-connected power plants. As per para. 14 of AMS-III.G, the baseline scenario is the situation where, in the absence of the project activity, biomass and other organic matter are left to decay within the project boundary, and methane is emitted to the atmosphere.

As per para. 19 of AMS-I.D, the baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

Therefore, it was verified that the baseline scenario of the project activity is LFG from Yueyang County MSW Landfill is emitted directly to the atmosphere and the equivalent amount of power would have been generated by the operation of grid-connected power plants and by the addition of new generation sources into CCPG.

The assessment team confirms that the Project has been approved by Chinese government by checking the Project approval /16/ and Environmental Impact Assessment (EIA) approval /18/. By checking laws and regulation, it is confirmed that the project activity is in complicate with all laws and regulations in China.

Besides, as per CDM Validation and Verification Standard for project activities version 03.0, "where the baseline scenario is not prescribed in the approved methodology, the DOE shall assess the list of identified credible alternatives to the project activity in the joint PD & MR selected to determine the most realistic baseline scenario." As the selected small-scale methodology clearly mention the baseline scenario and the same has been opted in this project, therefore, no further analysis on baseline is required.

In conclusion, the assessment team confirms that the joint PD & MR conforms to the guidance given by EB via CDM Validation and Verification Standard for project activities version 03.0 and VCS via VCS standard Version 4.4.

3.4.5 Additionality

As per the Methodological Tool "Positive lists of technologies", the project activities at new or existing landfills (greenfield or brownfield) are deemed automatically additional, if it is demonstrated that prior to the implementation of the project activities the landfill gas (LFG) was only vented and/or flared (in the case of brownfield projects) or would have been only vented and/or flared (in the case of greenfield projects) but not utilized for energy generation, and that under the project activities any of the following conditions are met:

- (a) The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW;
- (b) The LFG is used to generate heat for internal or external consumption;
- (c) The LFG is flared.

The project activity uses LFG which is collected from Yueyang County MSW Landfill Site to generate electricity. Prior to the implementation of the project activity, the LFG from Yueyang County MSW Landfill Site was only vented, not utilized for energy generation.

By interviewing with local stakeholders, staff from local government, staff from Yueyang County MSW Landfill Site and checking website of People's Government of Yueyang County (<http://www.yyx.gov.cn/>), Yueyang Municipal People's Government (<http://www.yueyang.gov.cn/index.html>), Yueyang Ecology and

Environment Bureau (<http://hbj.yueyang.gov.cn/index.htm>), it is confirmed by the assessment team that only the project activity uses LFG from Yueyang County MSW Landfill Site to generate electricity.

By checking Project Approval /16/ and EIA Approval /18/, the total installed capacity of the project activity is 2.134MW, which is below 10 MW. Therefore, the project activity is deemed automatically additional.

3.4.6 Quantification of GHG Emission Reductions and Removals

The assessment team checked the baseline, project and leakage calculation and confirm that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detail analysis is as below:

Baseline emissions

Since the project activity utilizes LFG generated from landfill site for electricity generation, baseline emissions are determined according to the following equation and comprise the following sources:

- (a) Methane emissions from the SWDS in the absence of the project activity;
- (b) Electricity generation using fossil fuels or supplied by the grid in the absence of the project activity;

$$BE_y = BE_{CH_4,y} + BE_{EC,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂e/yr)

$BE_{CH_4,y}$ = Baseline emissions of methane from the SWDS in year y (tCO₂e/yr)

$BE_{EC,y}$ = Baseline emissions associated with electricity generation in year y (tCO₂/yr)

● Baseline emissions of methane from the SWDS ($BE_{CH_4,y}$)

As per AMS-III.G, Baseline emissions of methane from the SWDS ($BE_{CH_4,y}$) is determined as follows:

$$BE_{CH_4,y} = \eta_{PJ} \times BE_{CH_4,SWDS,y} - (1 - OX) \times F_{CH_4,BL,y} \times GWP_{CH_4}$$

Where:

$BE_{CH_4,y}$ = Baseline emissions of methane from the SWDS in year y (tCO₂e/yr)

$BE_{CH_4,SWDS,y}$ = Methane emission potential of a solid waste disposal site (in t CO₂e), calculated using the methodological tool “Emissions from solid waste disposal sites”. This tool may be used:

- With the factor “f=0.0” because the amount of LFG that would have been captured and destroyed is already accounted for in this equation;
- With the definition of year x as ‘the year since the landfill started receiving wastes, x runs from the first year of landfill operation (x=1) to the year for which emissions are calculated (x=y)’.

The amount of waste type j deposited each year x ($W_{j,x}$) shall be determined by sampling (as specified in the above-mentioned tool), in the case that waste is generated during the crediting period. Alternatively, for existing SWDS, if the pre-existing amount and composition of the wastes in the landfill are unknown, they can be estimated by using parameters related to the serviced population or industrial activity, or by comparison with other landfills with similar conditions at regional or national level.

- OX = Oxidation factor (reflecting the amount of methane from SWDS that is oxidised in the soil or other material covering the waste) (dimensionless). A default value of 0.1 may be used
- η_{PJ} = Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex ante estimation only. A default value of 50 per cent may be used.
- $F_{CH_4,BL,y}$ = Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y (t CH₄). The relevant procedures in "ACM0001: Flaring or use of landfill gas" may be followed, as well as taking into account the compliance with the relevant local laws and regulation if such laws and regulations exist.
- GWP_{CH_4} = Global Warming Potential of CH₄ (tCO₂e/tCH₄)

$BE_{CH_4,SWDS,y}$ is determined using the methodological tool "Emissions from solid waste disposal sites". The following guidance should be taken into account when applying the tool:

- f_y in the tool shall be assigned a value of 0 because the amount of LFG that would have been captured and destroyed is already accounted for in equation (2) of methodology AMS-III.G;
- In the tool, x begins with the year that the SWDS started receiving wastes (e.g. the first year of SWDS operation); and
- Sampling to determine the fractions of different waste types is not necessary because the waste composition can be obtained from previous studies.

For the project, the amount of methane generated from disposal of waste at the SWDS for year y ($BE_{CH_4,SWDS,y}$) is calculated using first order decay (FOD) model as follows:

$$BE_{CH_4,SWDS,y} = \phi_y \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y \times \sum_{x=1}^y \sum_j (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{-k_j}))$$

Where:

- $BE_{CH_4,SWDS,y}$ = Baseline, project or leakage methane emissions occurring in year y generated from waste disposal at a SWDS during a time period ending in year y (tCO₂e/yr)
- x = Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period ($x = 1$) to year y ($x = y$)

y	=	Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)
$DOC_{f,y}$	=	Fraction of degradable organic carbon (DOC) that can decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)
$W_{j,x}$	=	Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)
ϕ_y	=	Model correction factor to account for model uncertainties for year y
f_y	=	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
GWP_{CH_4}	=	Global Warming Potential of methane
OX	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
F	=	Fraction of methane in the SWDS gas (volume fraction)
MCF_y	=	Methane correction factor
DOC_j	=	Fraction of degradable organic carbon in the waste type j (weight fraction)
k_j	=	Decay rate for the waste type j (1 / yr)
j	=	Type of residual waste or types of waste in the MSW

Determination of $F_{CH_4,BL,y}$

This section provides a procedure to determine the amount of methane that would have been captured and destroyed (by flaring) in the baseline due to regulatory or contractual requirements, to address safety and odour concerns, or for other reasons (collectively referred to as requirement in this section). The four cases in the following table are distinguished. The appropriate case should be identified, and the corresponding instructions followed.

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 4	Yes	Yes

Case 1: No requirement to destroy methane exists and no existing LFG capture system

The assessment team confirmed that no regulation or standard that mandatory requires methane destruction in Yueyang County MSW Landfill Site. Prior to the implementation of the project activity, the LFG from Yueyang County MSW Landfill Site was only vented, while no LFG capture and destruction system exists. Therefore, it is confirmed that the project is satisfied with the situation of Case 1: No requirement to destroy methane exists and no existing LFG capture system.

The parameters required to apply the FOD model is determined as:

Parameter	Data unit	Value applied	Data source														
Φ_y	/	0.75	Default values from "Emissions from solid waste disposal sites" Application A														
OX	/	0.1	IPCC 2006 Guidelines for National Greenhouse Gas Inventories														
F	/	0.5	IPCC 2006 Guidelines for National Greenhouse Gas Inventories														
$DOC_{r,y}$	Weight fraction	0.5	IPCC 2006 Guidelines for National Greenhouse Gas Inventories														
MCF_y	/	1.0	Default values from "Emissions from solid waste disposal sites" Application A By checking management manual of Yueyang County MSW Landfill Site, the assessment team confirmed that waste was directed to specific deposition areas, a degree of control of scavenging and a degree of control of fires. Yueyang County MSW Landfill Site includes at least one of the following: (i) cover material; (ii) mechanical compacting; or (iii) levelling of the waste. Therefore, 1.0 for anaerobic managed solid waste disposal sites was applied for the project activity.														
k_j	1/yr	<table><tr><td>Pulp, paper and cardboard</td><td>0.06</td></tr><tr><td>Textiles</td><td>0.06</td></tr><tr><td>Food, food waste, beverages and tobacco</td><td>0.185</td></tr><tr><td>Wood and wood products</td><td>0.03</td></tr><tr><td>Garden, yard and park waste</td><td>0.10</td></tr><tr><td>Others</td><td>-</td></tr></table>	Pulp, paper and cardboard	0.06	Textiles	0.06	Food, food waste, beverages and tobacco	0.185	Wood and wood products	0.03	Garden, yard and park waste	0.10	Others	-	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 3.3)		
Pulp, paper and cardboard	0.06																
Textiles	0.06																
Food, food waste, beverages and tobacco	0.185																
Wood and wood products	0.03																
Garden, yard and park waste	0.10																
Others	-																
$P_{n,j,x}$ (Weight fraction of waste type j in the landfill)	1/yr	<table><tr><td>Pulp, paper and cardboard</td><td>4.30%</td></tr><tr><td>Textiles</td><td>7.30%</td></tr><tr><td>Food, food waste, beverages and tobacco</td><td>53.43%</td></tr><tr><td>Wood and wood products</td><td>4.90%</td></tr><tr><td>Garden, yard and park waste</td><td>7.10%</td></tr><tr><td>Rubber, plastic, Glass, metal and other inert</td><td>22.97%</td></tr><tr><td>Total</td><td>100%</td></tr></table>	Pulp, paper and cardboard	4.30%	Textiles	7.30%	Food, food waste, beverages and tobacco	53.43%	Wood and wood products	4.90%	Garden, yard and park waste	7.10%	Rubber, plastic, Glass, metal and other inert	22.97%	Total	100%	The waste composition was obtained from FSR of the project activity. This parameter was estimated once.
Pulp, paper and cardboard	4.30%																
Textiles	7.30%																
Food, food waste, beverages and tobacco	53.43%																
Wood and wood products	4.90%																
Garden, yard and park waste	7.10%																
Rubber, plastic, Glass, metal and other inert	22.97%																
Total	100%																

DOC _i	/	Pulp, paper and cardboard	40%	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Tables 2.4 and 2.5)
		Textiles	24%	
		Food, food waste, beverages and tobacco	15%	
		Wood and wood products	43%	
		Garden, yard and park waste	20%	
		Others	-	
f _y	/	0		Requirement from the tool "Emissions from solid waste disposal sites"
GWP _{CH4}	tCO ₂ e/tCH ₄	27.9		IPCC AR6

According to the methodological tool "Emissions from solid waste disposal sites" (version 08.1), the amount of methane generated from disposal of waste at the SWDS for year y (BE_{CH4,SWDS,y}) is calculated using equations mentioned above. However, the estimated methane amount in landfill gas of the project during the 10 years crediting period is higher than the methane amount that could be destroyed by the project with the capacity of 2.134MW. A theoretical calculation of methane destroyed by capacity of 2.134MW is ex-ante estimated using parameters from FSR, including annual operating hours (7200 hours), the power generation efficiency (40.2%), plant self-consumption rate (6%), methane calorific value (0.0504 TJ/tCH₄). The detailed calculation process in the ER calculation spreadsheet /2/ has been confirmed by the assessment team to be correct as follows:

Year	Methane amount estimated ex-ante (m ³)	η _{PJ}	Theoretical installed capacity (MW)	Actual installed capacity (MW)	Annual electricity generation (MWh)	Plant self-consumption rate (%)	Annual electricity supplied to the grid (MWh)	Annual methane destroyed based on annual electricity generation (tCH ₄)
2021	2,895,864	80%	1.297	2.134	9,335	6%	8,775	1,667
2022	2,755,825	80%	1.234	2.134	8,884	6%	8,351	1,587
2023	2,376,465	80%	1.064	2.134	7,661	6%	7,201	1,368
2024	2,056,067	80%	0.921	2.134	6,628	6%	6,230	1,184
2025	1,785,017	80%	0.799	2.134	5,754	6%	5,409	1,028
2026	1,555,299	80%	0.696	2.134	5,014	6%	4,713	896
2027	1,360,230	80%	0.609	2.134	4,385	6%	4,122	783
2028	1,194,230	80%	0.535	2.134	3,850	6%	3,619	688
2029	1,052,645	80%	0.471	2.134	3,393	6%	3,190	606
2030	931,586	80%	0.417	2.134	3,003	6%	2,823	536
2031	827,804	80%	0.371	2.134	2,669	6%	2,508	477

Base on the above parameters and equations, the baseline emissions of methane from the SWDS (BE_{CH4,y}) is calculated as:

Period	F _{CH4,EL,y}	F _{CH4,PJ,y}	OX _{top-layer}	F _{CH4,BL,y}	BE _{CH4,y}
	tCH ₄	tCH ₄	Dimensionless	tCH ₄	tCO ₂ e
26/11/2021 to 31/12/2021	164	164	0.1	0	4,564
01/01/2022 to 31/12/2022	1,587	1,587	0.1	0	44,041
01/01/2023 to 31/12/2023	1,368	1,368	0.1	0	37,978

01/01/2024 to 31/12/2024	1,184	1,184	0.1	0	32,858
01/01/2025 to 31/12/2025	1,028	1,028	0.1	0	28,526
01/01/2026 to 31/12/2026	896	896	0.1	0	24,855
01/01/2027 to 31/12/2027	783	783	0.1	0	21,737
01/01/2028 to 31/12/2028	688	688	0.1	0	19,085
01/01/2029 to 31/12/2029	606	606	0.1	0	16,822
01/01/2030 to 31/12/2030	536	536	0.1	0	14,887
01/01/2031 to 25/11/2031	430	430	0.1	0	11,924
Total					257,277

● **Baseline emissions associated with electricity generation ($BE_{EC,y}$)**

As per AMS-I.D (version 18.0), the baseline emissions are to be calculated as follows:

$$BE_{EC,y} = EG_{PJ,y} \times EF_{grid,y}$$

Where:

$BE_{EC,y}$ = Baseline emissions associated with electricity generation in year y (t CO₂)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO₂/MWh)

If the project activity is the installation of a greenfield power plant, then:

$$EG_{PJ,y} = EG_{PJ,facility,y}$$

Where:

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)

As per the "Tool to calculate the emission factor of an electricity system", the grid emission factor ($EF_{grid,y} = EF_{grid,CM,y}$) is calculated as the weighted average of the operating margin (0.5) and build margin (0.5) values. The value of combined margin is sourced from 2019 Baseline Emission Factors for Regional Power Grids in China dated 29/12/2020 published by China DNA. /22/ calculates the data based on "Tool to Calculate the Emission Factor for an Electricity System", Version 07.0. No further assessment is required for grid emission calculation as the ex-ante value is sourced directly from the Chinese DNA. The grid emission factor is fixed ex-ante for the crediting period, which is calculated as:

$$EF_{grid,y} = EF_{grid,CM,y} = 0.8587 \times 0.5 + 0.2854 \times 0.5 = 0.5720 \text{ tCO}_2/\text{MWh}$$

Parameter	EG _{PJ, facility, y}	EF _{grid, CM, y}	BE _{EC, y}
Unit	MWh	tCO ₂ e/MWh	tCO ₂ e
26/11/2021 to 31/12/2021	866	0.5720	495
01/01/2022 to 31/12/2022	8,351	0.5720	4,776
01/01/2023 to 31/12/2023	7,201	0.5720	4,119
01/01/2024 to 31/12/2024	6,230	0.5720	3,563
01/01/2025 to 31/12/2025	5,409	0.5720	3,094
01/01/2026 to 31/12/2026	4,713	0.5720	2,695
01/01/2027 to 31/12/2027	4,122	0.5720	2,357
01/01/2028 to 31/12/2028	3,619	0.5720	2,069
01/01/2029 to 31/12/2029	3,190	0.5720	1,824
01/01/2030 to 31/12/2030	2,823	0.5720	1,614
01/01/2031 to 25/11/2031	2,261	0.5720	1,293
Total	48,785	/	27,899

The baseline emission of the project activity is calculated as:

Period	BE _{CH₄, y}	BE _{EC, y}	BE _y
	tCO ₂ e	tCO ₂ e	tCO ₂ e
26/11/2021 to 31/12/2021	4,564	495	5,059
01/01/2022 to 31/12/2022	44,041	4,776	48,817
01/01/2023 to 31/12/2023	37,978	4,119	42,097
01/01/2024 to 31/12/2024	32,858	3,563	36,421
01/01/2025 to 31/12/2025	28,526	3,094	31,620
01/01/2026 to 31/12/2026	24,855	2,695	27,550
01/01/2027 to 31/12/2027	21,737	2,357	24,094
01/01/2028 to 31/12/2028	19,085	2,069	21,154
01/01/2029 to 31/12/2029	16,822	1,824	18,646
01/01/2030 to 31/12/2030	14,887	1,614	16,501
01/01/2031 to 25/11/2031	11,924	1,293	13,217
Total (tones of CO₂e)	257,277	27,899	285,176

Project emissions

As per AMS-III.G. (Version 10.0), Project emissions are calculated as follows:

$$PE_y = PE_{Power, y} + PE_{flare, y} + PE_{process, y}$$

Where:

$$PE_y = \text{Project emissions in year } y \text{ (t CO}_2\text{e)}$$

$PE_{Power,y}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO₂e)

$PE_{flare,y}$ = Emissions from flaring or combustion of the landfill gas stream in the year y (t CO₂e)

$PE_{process,y}$ = Emissions from the landfill gas upgrading process in the year y (t CO₂e), determined by following the relevant procedures described in annex 1 of AMS-III.H.

By checking FSR /15/ and through on-site interview, the assessment team confirmed that there is no fossil fuel consumption in the project activity. The electricity for the operation of the installed facilities has been considering when calculating parameter $EG_{PJ, facility,y}$, which is consistent with AMS-I.D. Therefore, $PE_{Power,y} = 0$.

Through site interview with staff from Yueyang County MSW Landfill Site operation entity, the assessment team confirmed that the flare will only be involved when the electricity generation system is not in operation. As confirmed with PP, VERs from flare will not be claimed, which is conservative. Under this situation, no baseline emission from flaring will be calculated and it is reasonable project emissions from flaring are excluded. Therefore, $PE_{flare,y} = 0$.

By checking FSR /15/ and through on-site interview, the assessment team confirmed that the project activity does not involve landfill gas upgrading process, Therefore, $PE_{process,y} = 0$.

Therefore, $PE_y = 0$.

Leakage

As per AMS-III.G (Version 10.0) and AMS-I.D (Version 18.0), the project does not need to consider leakage. Therefore, $LE_y = 0$

Emission Reductions

Emission reductions are calculated as follows:

$$ER_{y,estimated} = BE_y - PE_y - LE_y$$

Hence for the project activity, the estimated amount of GHG emission reductions (ER_y) is 285,176 tCO₂e during the crediting period from 26/11/2021 to 25/11/2031, resulting in estimated average annual emission reductions of 28,517 tCO₂e.

Year	BE _y (tCO ₂ e)	PE _y (tCO ₂ e)	LE _y (tCO ₂ e)	ER _y (tCO ₂ e)
26/11/2021 to 31/12/2021	5,059	0	0	5,059
01/01/2022 to 31/12/2022	48,817	0	0	48,817
01/01/2023 to 31/12/2023	42,097	0	0	42,097
01/01/2024 to 31/12/2024	36,421	0	0	36,421
01/01/2025 to 31/12/2025	31,620	0	0	31,620

01/01/2026 to 31/12/2026	27,550	0	0	27,550
01/01/2027 to 31/12/2027	24,094	0	0	24,094
01/01/2028 to 31/12/2028	21,154	0	0	21,154
01/01/2029 to 31/12/2029	18,646	0	0	18,646
01/01/2030 to 31/12/2030	16,501	0	0	16,501
01/01/2031 to 25/11/2031	13,217	0	0	13,217
Total	285,176	0	0	285,176

Furthermore, as per AMS-III.G. (Version 10.0), the **actual emission reduction achieved** by the project activity during the crediting period will be calculated using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

$$ER_{y,calculated} = (1 - OX) \times (F_{CH_4,PJ,y} - F_{CH_4,BL,y}) \times GWP_{CH_4} - PE_y - LE_y$$

Where:

$F_{CH_4,PJ,y}$ = Methane captured and destroyed/gainfully used by the project activity in the year y (tCH₄)

$$F_{CH_4,PJ,y} = D_{CH_4,y} \times w_{CH_4,y} \times \sum_i LFG_{i,y}$$

Where:

$LFG_{i,y}$ = Landfill gas destroyed via method i (flaring, fuelling, combustion, injection to a grid, etc.) in year y (m³LFG). The flow or volume measurement shall be made either on a dry basis or at the same humidity as $w_{CH_4,y}$

$w_{CH_4,y}$ = Methane content in landfill gas in year y (volume fraction, m³CH₄/m³LFG). Landfill gas composition shall be measured either on a dry basis or at the same humidity as used to determine $LFG_{i,y}$

$D_{CH_4,y}$ = Density of methane at the temperature and pressure of the landfill gas in year y (tonnes/m³). If $LFG_{i,y}$ is reported at normal conditions of temperature and pressure, the density of methane is also determined at normal conditions

For project activities that utilize the recovered methane for power generation, $F_{CH_4,PJ,y}$ may be calculated as follows, based on the amount of monitored electricity generation, **without monitoring methane flow and concentration**:

$$F_{CH_4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH_4} \times EE_y} \times D_{CH_4}$$

Where:

EG_y	=	Electricity generation in year y (MWh)
3600	=	Conversion factor (1 MWh = 3600 MJ)
NCV_{CH_4}	=	NCV of methane (MJ/Nm ³) use default value: 35.9 MJ/Nm ³
EE_y	=	Energy Conversion Efficiency of the project equipment determined from one of the following options: • Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation • Default efficiency of 40 per cent

The assessment team confirms that all data sources and assumptions are appropriate and calculations are correct, applicable to the project activity and will result in a conservative estimate of the emission reductions. In general, the assessment team is able to confirm the following:

- All assumptions and data used by the project participants are listed in the joint PD & MR (version 3.0 dated 08/10/2023) and/or supporting documents, including their references and sources;
- All documentation used by the project participants as the basis for assumptions and source of data is correctly quoted and interpreted in the joint PD & MR (version 3.0 dated 08/10/2023);
- All values used in joint PD & MR (version 3.0 dated 08/10/2023) are considered reasonable in the context of the project activity;
- The baseline methodology has been applied correctly to calculate project emissions, baseline emissions, and leakage emissions;
- All estimates of the baseline, project and leakage emissions can be replicated using the data and parameter values provided in joint PD & MR (version 3.0 dated 08/10/2023).

3.4.7 Methodology Deviations

As plant specific fuel consumption and electricity generation data are not publicly available in China, the Guidance caused by DNV's request for deviation of Chinese project activities for the CDM baseline methodology AM0005¹ has been applied for calculating the build margin (BM) emission factor of this project activity. The deviation has no impact on conservativeness of the quantification of GHG emission reductions or removals.

Except this, no methodology deviation is applied in the project.

¹ http://cdm.unfccc.int/UserManagement/FileStorage/AM_CLAR_QEJWJEF3CFBP10ZAK6V5YXPQKK7WYJ

3.4.8 Monitoring Plan

The project applies the approved monitoring methodology AMS-III.G: Landfill methane recovery, Version 10.0. The selected monitoring methodology is applicable for the project. The monitoring plan is in accordance with the monitoring methodologies. The monitoring plan will give opportunity for real measurements of achieved emission reductions and contains principles and concepts on which it is based, operational and monitoring obligations of the project owner like resources involved in the monitoring process, training, support activities, calibration and data collection, quality assurance procedures, data management, electronic support tools.

Parameters determined ex-ante

Data / Parameter:	GWP_{CH_4}
Data unit:	/
Description:	Global warming potential of CH_4
Source of data used:	IPCC AR6
Value applied:	27.9

Data / Parameter:	η_{PJ}
Data unit:	Dimensionless
Description:	Efficiency of the LFG capture system that will be installed in the project activity
Source of data used:	Feasibility Study Report (FSR)
Value applied:	80%

Data / Parameter:	D_{CH_4}
Data unit:	$tCH_4/m^3 CH_4$
Description:	Density of methane at the temperature and pressure of the landfill gas in the year y
Source of data used:	Default value as per the latest version of "Project emissions from flaring"
Value applied:	0.716 at normal condition

Data / Parameter:	Φ_y
-------------------	----------

Data unit:	Dimensionless		
Description:	Efficiency of the LFG capture system that will be installed in the project activity		
Source of data used:	Emissions from solid waste disposal sites (version 08.1)		
Value applied:	0.75		
		Humid/wet conditions	Dry conditions
	Application A	0.75	0.75
	Application B	0.85	0.80
	The project selects Option 1 in tool “Emissions from solid waste disposal sites” (version 08.1): Use a default value to determine ϕ_y , and the project belongs to the situation of Application A: The VCS project activity mitigates methane emissions from a specific existing SWDS, therefore 0.75 is chosen for ϕ_y .		

Data / Parameter:	f_y
Data unit:	/
Description:	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
Source of data used:	Tool “Emissions from solid waste disposal sites” Select the maximum value from the following: (a) contract or regulation requirements specifying the amount of methane that must be destroyed/used (if available) and (b) historic data on the amount captured
Value applied:	No contract or regulation requirements of LFG recovery and utilization were identified by the assessment team. By interviewing with staff from landfill site, it is confirmed no LFG recovery and utilization facility was in the baseline scenario. Hence f_y is 0.

Data / Parameter:	NCV_{CH_4}
Data unit:	MJ/Nm ³
Description:	Global warming potential of CH ₄
Source of data used:	AMS-III.G. (version 10.0).
Value applied:	35.9

Data / Parameter:	OX
Data unit:	/
Description:	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
Source of data used:	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied:	0.1 Through site interview and inspection, the assessment team confirmed that Yueyang County MSW Landfill Site is covered with plastic film or soil, therefore, 0.1 was chosen for conservative.

Data / Parameter:	F
Data unit:	/
Description:	Fraction of methane in the SWDS gas (volume fraction)
Source of data used:	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied:	0.5

Data / Parameter:	DOC_{r,y}
Data unit:	Weight fraction
Description:	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS
Source of data used:	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied:	0.5

Data / Parameter:	MCF_y
Data unit:	/
Description:	Methane correction factor
Source of data used:	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied:	Yueyang County MSW Landfill Site is an anaerobic managed solid waste disposal site. As per IPCC 2006 Guidelines for National Greenhouse Gas Inventories, MCF _y = 1.0

Data / Parameter:	W_{j,x}																																				
Data unit:	t																																				
Description:	Amount of organic waste type j disposed in the SWDS in the year x																																				
Source of data used:	Estimated once in the Feasibility study report (FSR)																																				
Value applied:	By checking FSR /15/, it was verified the amount of organic waste disposed in the SWDS were: <table border="1"> <thead> <tr> <th>Year</th><th>W_{j,x} (annual disposed waste) (t)</th></tr> </thead> <tbody> <tr><td>2013</td><td>8,769</td></tr> <tr><td>2014</td><td>35,223</td></tr> <tr><td>2015</td><td>32,497</td></tr> <tr><td>2016</td><td>36,096</td></tr> <tr><td>2017</td><td>205,424</td></tr> <tr><td>2018</td><td>139,462</td></tr> <tr><td>2019</td><td>107,983</td></tr> <tr><td>2020</td><td>107,736</td></tr> <tr><td>2021</td><td>110,030</td></tr> <tr><td>2022</td><td>47,931</td></tr> </tbody> </table> <table border="1"> <thead> <tr> <th>Waste type of Landfill site</th><th>Weight content (%)</th></tr> </thead> <tbody> <tr><td>Wood and wood products</td><td>4.90%</td></tr> <tr><td>Pulp, paper and cardboard</td><td>4.30%</td></tr> <tr><td>Food, food waste, beverages and tobacco</td><td>53.43%</td></tr> <tr><td>Textiles</td><td>7.30%</td></tr> <tr><td>Garden, yard and park waste</td><td>7.10%</td></tr> <tr><td>Glass, plastic, metal, other inert waste</td><td>22.97%</td></tr> </tbody> </table>	Year	W _{j,x} (annual disposed waste) (t)	2013	8,769	2014	35,223	2015	32,497	2016	36,096	2017	205,424	2018	139,462	2019	107,983	2020	107,736	2021	110,030	2022	47,931	Waste type of Landfill site	Weight content (%)	Wood and wood products	4.90%	Pulp, paper and cardboard	4.30%	Food, food waste, beverages and tobacco	53.43%	Textiles	7.30%	Garden, yard and park waste	7.10%	Glass, plastic, metal, other inert waste	22.97%
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Data / Parameter:	DOC_j
Data unit:	/
Description:	Fraction of degradable organic carbon in the waste type j (weight fraction)
Source of data used:	TOOL04: "Emissions from solid waste disposal sites" (version 08.1)
Value applied:	For MSW, the following values for the different waste types j should be

	applied:														
	<table><tr><th>Waste classification</th><th>DOC_j (% wet waste)</th></tr><tr><td>Wood and wood product</td><td>43</td></tr><tr><td>Pulp, paper and cardboard (other than sludge)</td><td>40</td></tr><tr><td>Food, food waste, beverage and tobacco (other than sludge)</td><td>15</td></tr><tr><td>Textiles</td><td>24</td></tr><tr><td>Gardens, yard and park was</td><td>20</td></tr><tr><td>Glass, metal and other inert waste</td><td>0</td></tr></table>	Waste classification	DOC _j (% wet waste)	Wood and wood product	43	Pulp, paper and cardboard (other than sludge)	40	Food, food waste, beverage and tobacco (other than sludge)	15	Textiles	24	Gardens, yard and park was	20	Glass, metal and other inert waste	0
	Waste classification	DOC _j (% wet waste)													
	Wood and wood product	43													
	Pulp, paper and cardboard (other than sludge)	40													
	Food, food waste, beverage and tobacco (other than sludge)	15													
	Textiles	24													
	Gardens, yard and park was	20													
Glass, metal and other inert waste	0														

Data / Parameter:	k _j					
Data unit:	1/yr					
Description:	Decay rate for the waste type j					
Source of data used:	TOOL04: “Emissions from solid waste disposal sites” (version 08.1), IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adopted from Volume 5, Table 3.3)					
Value applied:	Apply the following default values for the different waste types j:					
	Waste type <i>j</i>		Boreal and Temperate (MAT≤ 20°C)		Tropical (MAT>20°C)	
			Dry (MAP/PET<1)	Wet (MAP/PET>1)	Dry (MAP <1000mm)	Wet (MAP >1000mm)
	Slowly degrading	Pulp, paper, cardboard (other than sludge), textiles	0.04	0.06	0.045	0.07
		Wood, wood products and straw	0.02	0.03	0.025	0.035
	Moderately degrading	Other (non-food) organic putrescible garden and park waste	0.05	0.10	0.065	0.17
	Rapidly degrading	Food, food waste, sewage sludge, beverages and	0.06	0.185	0.085	0.40

	tobacco				
As per public statistics data /24/, the climate data for the location of Yueyang County MSW Landfill Site is: MAT – Mean annual temperature: 17°C MAP – Mean annual precipitation: 1320.3 mm PET – the potential evapotranspiration: 1,013mm Therefore, the value of k_j under tropical ($MAT \leq 20^\circ C$), wet ($MAP/PET > 1$) is applied for the project activity.					

Data / Parameter:	EF_{grid,OM,y}
Data unit:	tCO ₂ /MWh
Description:	Operating margin CO ₂ emission factor in year y
Source of data used:	2019 Baseline Emission Factors for Regional Power Grids in China dated 29/12/2020 published by China DNA
Value applied:	0.8587

Data / Parameter:	EF_{grid,BM,y}
Data unit:	tCO ₂ /MWh
Description:	Build margin CO ₂ emission factor in year y
Source of data used:	2019 Baseline Emission Factors for Regional Power Grids in China dated 29/12/2020 published by China DNA
Value applied:	0.2854

Data / Parameter:	EF_{grid,CM,y} (EF_{grid,y})
Data unit:	tCO ₂ /MWh
Description:	Combined margin CO ₂ emission factor in year y
Source of data used:	Calculated as per “Tool to calculate the emission factor for an electricity system” Version 07.0
Value applied:	0.5720

Data / Parameter:	EE _y
Data unit:	%
Description:	Energy Conversion Efficiency of the project equipment
Source of data used:	Specification provided by the equipment manufacturer
Value applied:	40.2%

Parameters determined ex-post

Parameter	Description	Monitoring procedure
EG _{PJ,facility,y}	Net amount of electricity generated using LFG by the project activity in year y	This parameter is calculated as difference between: the quantity of electricity supplied by the project to the grid (EG_{export,y}); and the quantity of electricity delivered to the project from the grid (EG_{import,y}). The formula is: $EG_{PJ,facility,y} = EG_{export,y} - EG_{import,y}$ The parameters EG_{export,y} and EG_{import,y} are continuously monitored by bi-directional electricity meter with accuracy not lower than 0.5S, installed at the outlet of the onsite substation. The meter readings are recorded regularly. The meter readings will be crosschecked with Electricity Receipts. The electricity meter was calibrated periodically as per relevant standards.
EG _y	Electricity generation by the project activity in year y	The parameters EG_y is continuously monitored by electricity meter with accuracy not lower than 0.5S was installed at the outlet of the gas generator. The meter readings are recorded regularly. The electricity meter was calibrated periodically as per relevant standards.

Based on the on-site visit and interviewed with staff from Yueyang Weiteng New Energy Co., Ltd., The assessment team confirmed that monitoring parameter has been correctly described in the joint PD & MR (version 3.0 dated 08/10/2023) and in compliance with the methodology AMS-III.G Version 10.0, CDM Validation and Verification Standard for project activities version 03.0, and VCS standard Version 4.4.

3.5 Non-Permanence Risk Analysis

Not applicable for the present project activity.

4 VERIFICATION FINDINGS

4.1 Accuracy of GHG Emission Reduction and Removal Calculations

4.1.1 Project Implementation Status

The selected monitoring period 26/11/2021 to 30/04/2023 is the 1st monitoring period of the project, which is within the crediting period 26/11/2021 to 25/11/2031.

The control system at the LFG power plant is automated and assures continuous operation, including monitoring on malfunction of equipment. By checking the daily operation and maintenance records and interview with staff from project owner, CTI can confirm that no serious malfunction happened in this monitoring period. As confirmed with PP, the project is unstable at the beginning of operation, especially from 26/11/2021 to 31/05/2022. By checking Operation Log /35/, it was verified no electricity was generated during the period from 01/09/2022 to 30/11/2022 resulted by temporary overhaul. As confirmed with PP and staff from the landfill site, it was verified that the unstable gas flow leading to unfavorable operation of the generator was resulted by high concentrated landfill leachate. And it is confirmed this temporary overhaul will not impact the project design and not lower the accuracy of emission reductions calculation.

On-site training for the related procedures including monitoring, recording and reporting was verified to be in place and their implementation was confirmed by interview with the key operators and observing the operation.

Through visual inspection and document review through site visit, CTI was able to confirm that:

- the project implementation is in accordance with the project description contained in the Joint PD & MR (version 3.0 dated 08/10/2023) and monitoring plan.
- all physical features of the proposed project activity including data collection systems and storage systems have been implemented in accordance with the Joint PD & MR (version 3.0 dated 08/10/2023) and monitoring plan.
- the monitoring plan in the Joint PD & MR (version 3.0 dated 08/10/2023) is in accordance with the approved methodology applied by the project activity, i.e. AMS-III.G (Version 10.0).
- The monitoring has been carried out in accordance with the monitoring plan contained in the Joint PD & MR (version 3.0 dated 08/10/2023).
- all parameters stated in the monitoring plan are monitored and reported appropriately.

All parameters required to be monitored by the monitoring plan as per the monitoring methodology AMS-III.G (Version 10.0) and the management system were assessed during the site visit. The joint PD & MR lists each parameter required by the monitoring plan and the information flow (i.e. from data generation, aggregation, recording, calculation and reporting) for these parameters is provided. The information flow for each parameter is further verified in the following sections.

4.1.2 Parameters monitored ex post

In this monitoring period, the parameters monitored ex post were justified as follows:

- **$EG_{PJ, facility, y}$: Net amount of electricity generated using LFG by the project activity in year y**

This parameter is calculated as difference between:

The quantity of electricity supplied by the project to the grid ($EG_{export, y}$); and the quantity of electricity delivered to the project from the grid ($EG_{import, y}$).

The formula is: $EG_{PJ, facility, y} = EG_{export, y} - EG_{import, y}$

Electricity meter M1 with accuracy of 0.5S was installed at the outlet of the onsite substation to continuously monitor $EG_{export, y}$ and $EG_{import, y}$. At 24:00 hr of last day of each month, the staff from project owner will record the electricity meter readings. Meanwhile, meter readings will remotely be transmitted to local power grid company and the data will be transcribed into Electricity Receipts.

The value of $EG_{PJ, facility, y}$ was sourced from Monthly Meter Reading Records (MRRs) /26/ issued by the project owner and Electricity Receipts /27/ /28/ issued by local power grid company covering the monitoring period. By checking operation log /35/ and as confirmed with the project owner, it was verified the two generators of the project was operating with no simultaneous shutdowns during July, August, and December 2022, and during January and February 2023. The electricity generated by the onsite generators is used for operation of the project activity and no electricity was imported from the grid. The data for MRRs and Electricity Receipts has been sent to the project participant for reporting of GHG emission reduction. The conservative one would be used for ER calculation. By checking the same, the assessment team confirmed that quantity of net electricity generation supplied by the project to the grid in this monitoring period were 5,675.868 MWh.

- **EG_y : Electricity generation by the project activity in year y**

Electricity meter M2 with accuracy 0.5S installed at the outlet of the gas generator to continuously monitor this parameter. At 24:00 hr of last day of each month, the staff from project owner will record the electricity meter readings in MRRs.

The value of EG_y was sourced from Monthly Meter Reading Records (MRRs) /26/ issued by the project owner covering the monitoring period. The same has been crosschecked with that in Monthly Electricity Data Proof issued by local power grid company /38/ and verified to be consistent. The data for MRRs /26/ and Monthly Electricity Data Proof issued by local power grid company /38/ has been sent to the project participant for reporting of GHG emission reduction. The assessment team confirmed that electricity generation in this monitoring period were 6,091.584 MWh.

Monitoring equipment and calibration

The monitoring equipments have been calibrated periodically as per the relevant industrial standard by the qualified third party to ensure the monitoring equipments' accuracy and in good conditions. The relevant information of monitoring equipments' calibration /29/ is listed as below.

Electric meter M1

Type	DSZ331
Serial No.	00000034426913
Accuracy	0.5S
Calibration frequency	Every two years
Date of calibration	10/10/2021
Date of Validity	09/10/2023
Electric meter M2	
Type	DTZ341
Serial No.	2212551926000344
Accuracy	0.5S
Calibration frequency	Every two years
Date of calibration	10/10/2021
Date of Validity	09/10/2023

The electricity meters were calibrated by a qualified third party, Power Supply Service Center of State Grid Hunan Electric Power Co., LTD. (Metering Center). Calibration records and accreditation certificates /29//32/ have been verified by the assessment team. By checking the industrial metering configuration standard “Technical Administrative Code of Electric Energy Metering (DL/T 448-2016)” /33/, CTI was able to confirm that the accuracy level of all meters meets the requirement stipulated in this standard. CTI can thus confirm that the accuracy of the meters is in line with relevant industry standard. By checking the calibration reports, CTI found the calibration frequency of these meters is every two years, which is in line with the calibrating standard “Technical Administrative Code of Electric Energy Metering (DL/T 448-2016)” /33/. Hence, CTI can confirm that the meters’ calibration frequency is in line with relevant industry standard, and the calibrations of meters are verified to be valid for the whole reporting period.

Data management and control

The project owner is responsible for operation and routine maintenance of power plant under the project activity. The quality assurance and quality control procedures have been addressed in the VER project management and monitoring manual /34/, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff and check records /34/-/35/ during on-site visit, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification. Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order..

4.1.3 Calculation of emission reductions

CTI confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

As per AMS-III.G. (Version 10.0) and AMS-I.D (Version 18.0),

$$BE_y = BE_{CH_4,y} + BE_{EC,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂e/yr)

$BE_{CH_4,y}$ = Baseline emissions of methane from the SWDS in year y (tCO₂e/yr)

$BE_{EC,y}$ = Baseline emissions associated with electricity generation in year y (tCO₂/yr)

As per AMS-III.G. (Version 10.0), the **actual emission reduction achieved** by the project activity during the crediting period will be calculated using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

$$ER_{y,calculated} = (1 - OX) \times (F_{CH_4,PJ,y} - F_{CH_4,BL,y}) \times GWP_{CH_4} - PE_y - LE_y$$

Where:

$F_{CH_4,PJ,y}$ = Methane captured and destroyed/gainfully used by the project activity in the year y (tCH₄)

$$F_{CH_4,PJ,y} = D_{CH_4,y} \times w_{CH_4,y} \times \sum_i LFG_{i,y}$$

Where:

$LFG_{i,y}$ = Landfill gas destroyed via method i (flaring, fuelling, combustion, injection to a grid, etc.) in year y (m³LFG). The flow or volume measurement shall be made either on a dry basis or at the same humidity as $w_{CH_4,y}$

$w_{CH_4,y}$ = Methane content in landfill gas in year y (volume fraction, m³CH₄/m³LFG). Landfill gas composition shall be measured either on a dry basis or at the same humidity as used to determine $LFG_{i,y}$

$D_{CH_4,y}$ = Density of methane at the temperature and pressure of the landfill gas in year y (tonnes/m³). If $LFG_{i,y}$ is reported at normal conditions of temperature and pressure, the density of methane is also determined at normal conditions

For project activities that utilize the recovered methane for power generation, $F_{CH_4,PJ,y}$ may be calculated as follows, based on the amount of monitored electricity generation, **without monitoring methane flow and concentration**:

$$F_{CH_4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH_4} \times EE_y} \times D_{CH_4}$$

Where:

EG_y = Electricity generation in year y (MWh)

3600 = Conversion factor (1 MWh = 3600 MJ)

NCV_{CH_4} = NCV of methane (MJ/Nm³) use default value: 35.9 MJ/Nm³

EE_y = Energy Conversion Efficiency of the project equipment determined from one of the following options:

- Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation
- Default efficiency of 40 per cent

The assessment team confirmed that no regulation or standard that mandatory requires methane destruction in Yueyang County MSW Landfill Site. Prior to the implementation of the project activity, the LFG from Yueyang County MSW Landfill Site was only vented, while no LFG capture and destruction system exists. Therefore, $F_{CH_4,BL,y}$ is 0.

Furthermore, during site visit, it is confirmed that the project activity is located at Yueyang County MSW Landfill Site in Weixing Village, Xinkai Town, Yueyang County, Yueyang City, Hunan province, People's Republic of China. Through site inspection, interviewing with staff from Yueyang County MSW Landfill Site and project owner, as well as checking FSR /15/, operation log /35/ of the project activity, it is confirmed that only the landfill gas recovered from Yueyang County MSW Landfill Site is used for power generation of the project activity, no other gas or fuels are used.

By checking Technical Specification of LFG generators /19/, the assessment team confirmed that the generators of the project activity uses electricity, not fuel, for start up. The same has been confirmed through checking operation log /35/ of the project activity.

Baseline emissions

- Baseline emissions of methane from the SWDS ($BE_{CH_4,y}$)

By checking data sources listed in the above table (section 4.1.1), based on the parameters monitored ex ante in section 3.4.8 and equations listed above, the assessment team confirmed the baseline emissions of methane from the SWDS ($BE_{CH_4,y}$) is calculated as:

Year	EG_y (MWh)	NCV_{CH_4} (MJ/Nm ³)	EE_y	D_{CH_4} (t/m ³)	$F_{CH_4,PJ,y}$ (tCH ₄)
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26/11/2021~30/11/2021	12.759	35.9	40.2%	0.000716	2
01/12/2021~31/12/2021	98.509	35.9	40.2%	0.000716	17
01/01/2022~31/01/2022	217.232	35.9	40.2%	0.000716	38
01/02/2022~28/02/2022	120.866	35.9	40.2%	0.000716	21
01/03/2022~31/03/2022	71.854	35.9	40.2%	0.000716	12
01/04/2022~30/04/2022	139.259	35.9	40.2%	0.000716	24
01/05/2022~31/05/2022	597.763	35.9	40.2%	0.000716	106
01/06/2022~30/06/2022	832.028	35.9	40.2%	0.000716	148
01/07/2022~31/07/2022	1013.623	35.9	40.2%	0.000716	181
01/08/2022~31/08/2022	883.820	35.9	40.2%	0.000716	157
01/09/2022~30/09/2022	0.000	35.9	40.2%	0.000716	0
01/10/2022~31/10/2022	0.000	35.9	40.2%	0.000716	0
01/11/2022~30/11/2022	0.000	35.9	40.2%	0.000716	0
01/12/2022~31/12/2022	262.396	35.9	40.2%	0.000716	46
01/01/2023~31/01/2023	646.172	35.9	40.2%	0.000716	115
01/02/2023~28/02/2023	493.547	35.9	40.2%	0.000716	88
01/03/2023~31/03/2023	340.492	35.9	40.2%	0.000716	60
01/04/2023~30/04/2023	361.264	35.9	40.2%	0.000716	64

BE_{CH₄,y} was calculated to be:

Year	OX	F _{CH₄,PJ,y} (tCH ₄)	F _{CH₄,BL,y} (tCH ₄)	GWP _{CH₄}	BE _{CH₄,y} (tCO ₂ e)
26/11/2021 to 31/12/2021	0.1	19	0	27.9	477
01/01/2022 to 31/12/2022	0.1	733	0	27.9	18,405
01/01/2023 to 30/04/2023	0.1	327	0	27.9	8,210
Total					27,092

Baseline emissions associated with electricity generation (BE_{EC,y})

Based on the values of monitoring parameters listed in table above, the baseline emissions associated with electricity generation (BE_{EC,y}) is calculated as:

Period	EG _{PJ,facility,y} (EG _{PJ,y}) (MWh)	EF _{grid,CM,y} (tCO ₂ e /MWh)	BE _{EC,y} (tCO ₂ e)
26/11/2021~30/11/2021	10.470	0.5720	5.99
01/12/2021~31/12/2021	84.335	0.5720	48.24
Sum 2021	94.805	0.5720	54
01/01/2022~31/01/2022	197.466	0.5720	112.95
01/02/2022~28/02/2022	103.181	0.5720	59.02
01/03/2022~31/03/2022	57.127	0.5720	32.68

01/04/2022~30/04/2022	118.318	0.5720	67.68
01/05/2022~31/05/2022	558.835	0.5720	319.65
01/06/2022~30/06/2022	781.132	0.5720	446.81
01/07/2022~31/07/2022	953.580	0.5720	545.45
01/08/2022~31/08/2022	833.790	0.5720	476.93
01/09/2022~30/09/2022	0.000	0.5720	0
01/10/2022~31/10/2022	0.000	0.5720	0
01/11/2022~30/11/2022	0.000	0.5720	0
01/12/2022~31/12/2022	246.600	0.5720	141.06
Sum 2022	3,850.029	0.5720	2,202
01/01/2023~31/01/2023	609.480	0.5720	348.62
01/02/2023~28/02/2023	467.490	0.5720	267.4
01/03/2023~31/03/2023	315.462	0.5720	180.44
01/04/2023~30/04/2023	338.602	0.5720	193.68
Sum 2023	1,731.034	0.5720	990
Total	5,675.868	0.5720	3,246

Therefore, the baseline emissions of the project activity were:

Year	BE _{CH₄,y} (tCO ₂ e)	BE _{EC,y} (tCO ₂ e)	BE _y (tCO ₂ e)
26/11/2021 to 31/12/2021	477	54	531
01/01/2022 to 31/12/2022	18,405	2,202	20,607
01/01/2023 to 30/04/2023	8,210	990	9,200
Total	27,092	3,246	30,338

Project emissions

As per AMS-III.G. (Version 10.0), Project emissions are calculated as follows:

$$PE_y = PE_{Power,y} + PE_{flare,y} + PE_{process,y}$$

Where:

PE_y = Project emissions in year y (t CO₂e)

$PE_{Power,y}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO₂e)

$PE_{flare,y}$ = Emissions from flaring or combustion of the landfill gas stream in the year y (t CO₂e)

$PE_{process,y}$ = Emissions from the landfill gas upgrading process in the year y (t CO₂e), determined

by following the relevant procedures described in annex 1 of AMS-III.H.

By checking FSR /15/ and through on-site interview, the assessment team confirmed that there is no fossil fuel consumption in the project activity. The electricity for the operation of the installed facilities has been considering when calculating parameter $EG_{PJ, facility, y}$, which is consistent with AMS-I.D. Therefore, $PE_{power, y} = 0$.

Through site interview with staff from Yueyang County MSW Landfill Site operation entity, the assessment team confirmed that the flare will only be involved when the electricity generation system is not in operation. As confirmed with PP, VERs from flare will not be claimed, which is conservative. Under this situation, no baseline emission from flaring will be calculated and it is reasonable project emissions from flaring are excluded. Therefore, $PE_{flare, y} = 0$.

By checking FSR /15/ and through on-site interview, the assessment team confirmed that the project activity does not involve landfill gas upgrading process, Therefore, $PE_{process, y} = 0$.

Therefore, $PE_y = 0$.

Leakage

As per AMS-III.G (Version 10.0) and AMS-I.D. (Version 18.0), the project does not need to consider leakage. Therefore, $LE = 0$.

Comparison of actual emission reductions or net anthropogenic GHG removals by sinks with estimates ex-ante

The emission reductions claimed are 30,338 tCO₂e in this monitoring period. Compared with expected emission reductions 67,716 tCO₂e (5,059 tCO₂e + 48,817 tCO₂e + 42,097 tCO₂e/365days*120days) calculated ex-ante, the reported emission reductions in this monitoring period are 55.20% less than that expected, which is considered to be in the reasonable variation range.

Verified GHG emission reductions and removals from 26/11/2021 to 30/04/2023:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
26/11/2021~31/12/2021	531	0	0	531
01/01/2022~01/12/2022	20,607	0	0	20,607
01/01/2023~30/04/2023	9,200	0	0	9,200
Total	30,338	0	0	30,338

Year	Ex-ante	Achieved	Percent	Justification for the difference
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	emissions reductions/ removals	emissions reductions/ removals	difference	
26/11/2021~30/04/2023	67,716	30,338	-55.20%	As confirmed with PP, the project is unstable at the beginning of operation, especially from 26/11/2021 to 31/05/2022. By checking Operation Log /35/, it was verified no electricity was generated during the period from 01/09/2022 to 30/11/2022 resulted by temporary overhaul. As confirmed with PP and staff from the landfill site, it was verified that the unstable gas flow leading to unfavorable operation of the generator was resulted by high concentrated landfill leachate.
Total	67,716	30,338	-55.20%	

CTI is able to confirm that the actual emission reductions reported in this monitoring period are reasonable and appropriate. CTI verified the input data for calculating emission reductions and the calculating process, and confirmed the result were complete and transparent. And CTI is able to confirm that GHG emission reductions and removals have been quantified correctly in accordance with the monitoring plan and applied methodology for this verification period.

4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

As a result of verification of the ER calculation process, the assessment team confirmed that all the parameters required for the determination of the emission reductions have been included in the joint PD & MR /1/ and ER Calculation Spreadsheet /3/ of this monitoring period and are consistent with the applied methodology AMS-III.G: Landfill methane recovery, Version 10.0, AMS-I.D.: Grid connected renewable electricity generation, (Version 18.0) and the monitoring plan contained in the joint PD & MR. The parameters are complete in this monitoring period.

After verifying the reported figures with the raw data sources, it's confirmed that the values of the parameters from the raw data sources are consistent with those quoted in the ER Calculation Spreadsheet of this monitoring period and joint PD & MR. the verification process for the same has been clearly described above in section 4 of the report.

All necessary documentations are collected, referenced and aggregated, which is easily accessible in hard-copy or electronic format. Measurements are performed by calibrated equipment, and the key data can also be cross-checked via other sources, such as records, receipts and inventory data. No assumptions are used that have any material influence on reported emission reductions.

CTI concludes that during this monitoring period, the evidences for determination of emission reductions are sufficient and reasonable, and the calculation of emission reductions is reliable.

5 VALIDATION AND VERIFICATION CONCLUSION

CTI Certification Co., Ltd (CTI) has performed the joint validation and verification of the project activity “Yueyang LFG Power Generation Project” in China (VCS Project ID: 4349) for the monitoring period 26/11/2021 to 30/04/2023 within the fixed crediting period 26/11/2021 to 25/11/2031.

The review of the project design documentation and the subsequent follow-up interviews have provided CTI with sufficient evidence to determine the fulfilment of stated criteria.

In the course of the validation and verification, 2 Clarification Requests (CLs), 1 Corrective Action Requests (CAR) and no Forwarded Action Requests (FARs) was raised.

The review of the project design documentation and additional documents related to baseline and monitoring methodology and subsequent background investigation have provided the CTI with sufficient evidence to validate the fulfilment of the stated criteria. In detail the conclusions can be summarized as follows:

- the baseline scenario is correctly defined as per the applied methodology and relate tools;
- the project is additional;
- all data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature;
- a reasonable level of assurance has been applied.
- the monitoring plan in the validated PD is as per the applied methodology AMS-III.G Version 10.0 and AMS-I.D. Version 18.0;
- the project has been implemented and operated as per the joint PD & MR;
- the project complies with the validation criteria for projects set out in VCS Version 4.4;
- The calculation of the ex ante emission reductions is carried out in a transparent and conservative manner, so that the calculated emission reductions of 285,176 tCO₂e are most likely to be achieved within the 10-year fixed crediting period started from 26/11/2021.

It is CTI's responsibility to provide an independent validation statement on the Project's compliance with the VCS Program requirements as set out in the VCS Program Rules. The conclusions of this report show that the project, as it was described in the project description contained in the joint PD & MR, is in line with all criteria applicable for the validation against the VCS standard Version 4.4 without any qualifications or limitations.

The verification is based on the baseline and monitoring methodology AMS-III.G Version 10.0 and AMS-I.D. Version 18.0, and the joint PD & MR (version 3.0 dated 08/10/2023). The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the monitoring plan and the reporting of GHG emission reductions on the basis set out within the joint PD & MR. Our verification approach was based on the requirements as defined under the applicable VCS

Version 4.4 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the monitoring plan in joint PD & MR is as per the applied methodology;
- the monitoring complies with the monitoring plan in the joint PD & MR;
- the joint PD & MR and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Version 4.4 and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.

It is CTI's responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CTI planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

CTI does not assume any responsibility towards the issuance and utilization of the VCU's hereby verified and certified. Request for issuance of VCU's shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

In CTI's opinion the GHG emissions reductions of the "Yueyang LFG Power Generation Project" for the monitoring period 26/11/2021 to 30/04/2023 are fairly stated in the joint PD & MR (version 3.0 dated 08/10/2023). The GHG emission reductions were calculated correctly on the basis of the approved methodology AMS-III.G Version 10.0 and the monitoring plan contained in the joint PD & MR (version 3.0 dated 08/10/2023).

CTI can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CTI confirms the following statement:

Verification period: From 26/11/2021 to 30/04/2023

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline	Project	Leakage	Net GHG
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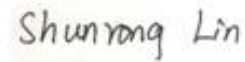
	emissions or removals (tCO ₂ e)	emissions or removals (tCO ₂ e)	emissions (tCO ₂ e)	emission reductions or removals (tCO ₂ e)
26/11/2021~31/12/2021	531	0	0	531
01/01/2022~01/12/2022	20,607	0	0	20,607
01/01/2023~30/04/2023	9,200	0	0	9,200
Total	30,338	0	0	30,338



Ms. Feng Tian

Team Leader

08/10/2023



Ms. Lin Shunrong

Technical Reviewer

08/10/2023

APPENDIX 1: ABBREVIATIONS

Abbreviations	Full texts
AFOLU	Agriculture, Forestry, and Other Land Use
BM	Build Margin
CDM	Clean Development Mechanism
CME	Coordinating/managing entity
CER	Certified Emission Reduction
CM	Combined Margin
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
DNA	Designated National Authority
DORs	Daily Operation Records
EB	Executive Board
GHG	Greenhouse gas(es)
GS	Gold Standard
EIA	Environmental Impact Assessment
ER	Emission Reduction
ETS	Emission Trading Scheme
FAR	Forward Action Request
GHG	Greenhouse gas(es)
IPCC	Intergovernmental Panel on Climate Change
MRRs	Monthly Meter Reading Records
MP	Monitoring Plan
PA	Project Activity
PD	Project Description
PP	Project Proponent
PPA	Power Purchase Agreement
QA/QC	Quality Assurance / Quality Control
UNFCCC	United Nations Framework Convention on Climate Change
VVS	Validation and Verification Standard
VCS	Verified Carbon Standard
VCSA	Verified Carbon Standard Association
VCU	Verified Carbon Unit
VVB	Validation/Verification Body

APPENDIX 2: <COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS>

Ms. Tian FENG

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

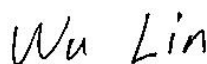
Qualification					
GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
√	√	√	√	-	-

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 8: Mining/mineral production	TA 8.1: Mining/mineral production
SS 13: Waste handling and disposal	TA 13.1: Solid waste and wastewater
	TA 13.2: Manure

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu LIN



Technical Competent Manager

Shenzhen, 25/10/2022

Ms. Shunrong LIN

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 3: Energy demand	TA 3.1: Energy demand
SS 13: Waste handling and disposal	TA 13.1: Solid waste and wastewater
	TA 13.2: Manure
SS 14: Afforestation and reforestation	TA 14.1: Afforestation and reforestation
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 25/10/2022

APPENDIX 3: <REFERENCE LIST>

1. Joint project design and monitoring report, version 1.1 dated 07/04/2023, version 2.0 dated 09/06/2023, version 3.0 dated 08/10/2023
2. ER calculation spreadsheet for ex-ante estimation
3. ER calculation spreadsheet for monitoring period from 26/11/2021 to 30/04/2023
4. VCS Standard Version 4.4, issued on 19/09/2019, updated on 17/01/2023
5. VCS Program Guide v4.3, issued on 19/09/2019, updated on 17/01/2023
6. CDM Validation and Verification Standard for project activities version 03.0
7. CDM Project Standard for project activities version 03.0
8. CDM project cycle procedure for project activities version 03.0
9. Approved methodology AMS-III.G: Landfill methane recovery, Version 10.0
10. Tool to calculate the emission factor for an electricity system, version 07.0
11. Emissions from solid waste disposal sites, version 08.1
12. Positive lists of technologies, Version 04.0
13. Grid connected renewable electricity generation, AMS-I.D., Version 18.0
14. Business License of Yueyang Weiteng New Energy Co., Ltd.
15. Feasibility Study Report (FSR) compiled by Xi'an Xujing Renewable Resources Co., Ltd on June 2020
16. Project Approval of the project activity issued by Yueyang County Development and Reform Bureau on 29/06/2020
17. Environmental Impact Assessment Report compiled by Hunan Huanmeida Environmental Protection Technology Co., Ltd. in Nov. 2020
18. Environmental Impact Assessment (EIA) Approval issued by Yueyang County Branch of Yueyang Ecology and Environment Bureau on 11/11/2020
19. Technical Specification of LFG generators
20. Nameplate of the equipment
21. Construction contract signed between the project owner and Yueyang City Daming Construction Co., Ltd on 02/12/2020
22. 2019 Baseline Emission Factors for Regional Power Grids in China dated 29/12/2020
23. Power Purchase Agreement signed between Yueyang Weiteng New Energy Co., Ltd. and local Power Grid Company

24. Meteorological data of Yueyang City:
http://www.yyx.gov.cn/37584/38154/38157/38160/38207/38211/40174/content_1314723.html
25. Operation Management Manual of Yueyang County MSW Landfill Site
26. Monthly Meter Reading Records (MRRs) provided by the project owner
27. Electricity Receipts for the quantity of electricity supplied by the project to the grid issued by local power grid company covering the monitoring period.
28. Electricity Receipts for the quantity of electricity delivered to the project from the grid issued by local power grid company covering the monitoring period.
29. Calibration Certificate of electricity meter M1 and M2 issued by Power Supply Service Center of State Grid Hunan Electric Power Co., LTD. (Metering Center) on 10/10/2021 (valid till 09/10/2023).
30. “Notice on Strengthening Enterprise Greenhouse Gas Emission Report Management” issued by National Development and Reform Committee of P.R.China on 28/03/2021
31. “List of key emissions in the national carbon emissions transaction quota management in 2019-2020” issued by Ministry of Ecology and Environment of P.R.China
32. Qualification of Power Supply Service Center of State Grid Hunan Electric Power Co., LTD. (Metering Center), issued by CNAS of the People's Republic of China issued on 14/08/2017, valid till 13/08/2023
33. Technical Administrative Code of Electric Energy Metering (DL/T 448-2016)
34. VCS monitoring manual
35. Operation log covering this monitoring period
36. Local stakeholder consultation questionnaire
37. China's 14th Five-Year Plan and the 2035 Vision Outline
http://www.gov.cn/xinwen/2021-03/13/content_5592681.htm
38. Monthly Electricity Data Proof issued by local power grid company
39. public information (lifetime of generator):
<https://news.bjx.com.cn/html/20141110/562392.shtml>

APPENDIX 4: <CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)>

CL ID	01	Section no.	3.1	Date: 30/05/2023
Description of CL				
Please supplement estimated net amount of electricity generated in section 1.1 of the PDMR.				
Project proponent response				Date: 09/06/2023
The estimated net amount of electricity generated has been updated according to the electricity receipts.				
Documentation provided by project proponent				
updated joint PD&MR				
DOE assessment				Date: 23/06/2023
The estimated net amount of electricity generated in section 1.1 of the PDMR, which has been verified by the assessment team to be correct. Therefore, CL01 was closed.				

CL ID	02	Section no.	4.1	Date: 30/05/2023
Description of CL				
As verified during site visit and checking PDMR, the project has no electricity generation during the period from 01/09/2022 to 30/11/2022 due to the temporary internal overhaul. Please further clarify the reasons resulted in the temporary internal overhaul.				
Project proponent response				Date: 09/06/2023
The reasons resulted in the temporary internal overhaul has been updated according to the production suspension statement.				
Documentation provided by project proponent				
updated joint PD&MR				
DOE assessment				Date: 23/06/2023

As confirmed with PP, the project is unstable at the beginning of operation, especially from 26/11/2021 to 31/05/2022. By checking Operation Log, it was verified no electricity was generated during the period from 01/09/2022 to 30/11/2022 resulted by temporary overhaul. As confirmed with PP and staff from the landfill site, it was verified that the unstable gas flow leading to unfavorable operation of the generator was resulted by high concentrated landfill leachate.

The reasons has been supplemented in the PDMR and therefore CL02 was closed.

CAR ID	01	Section no.	4.1	Date: 30/05/2023
Description of CL				
Please supplement information of electricity M2, including type. Serial No., accuracy, calibration frequency, date of calibration, date of calibration validity, in section 7.1 of the PDMR.				
Project proponent response				Date: 09/06/2023
The information of electricity M2, including type. Serial No., accuracy, calibration frequency, date of calibration, date of calibration validity, has been updated in section 7.1 of the PDMR, according to the nameplate information and calibration report of the electricity meter.				
Documentation provided by project proponent				
updated joint PD&MR				
DOE assessment				Date: 23/06/2023
The information of electricity M2, including type. Serial No., accuracy, calibration frequency, date of calibration, date of calibration validity, has been supplemented in section 7.1 of the PDMR and verified by the assessment team to be correct.				
Therefore, CAR01 was closed.				